

**Recurrent Vision Transformers as Models of Primate  
Visual Attention:  
A Comprehensive Architecture Study**

**Internal progress report**

RViT architecture program  
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## Abstract

This report consolidates everything the architecture program has learned about building a recurrent vision transformer (RViT) that reproduces the behavioural and neural signatures of primate covert spatial attention while learning end-to-end from sparse reward. Across roughly thirty network variants and two task horizons we isolate a small number of load-bearing findings. **(1) Perception is the bottleneck on the short (seven-step) task.** Sparse reward cannot teach the front-end to represent orientation: a reconstruction-pretrained VAE encoder lifts the linear decodability of Gabor orientation from  $R^2 = 0.24$  to  $R^2 = 0.97$ , and with that single change the exact paper network learns the task on the first attempt. **(2) The horizon governs which strategy is viable.** On the long task a weak front-end suffices because the change persists long enough to be detected by slow temporal integration (“optic flow”); the short task removes that escape hatch and forces accurate single-frame perception. **(3) Coupling between the policy and value pathways is a precondition for learning,** not an optimisation: cutting the shared recurrent memory collapses an otherwise identical split-pathway model to chance. **(4) The working model reproduces the paper’s core behavioural figures** — a reliability-scaled psychometric cueing benefit ( $+1.5^\circ \rightarrow +7.5^\circ$  across  $25\% \rightarrow 100\%$  validity) and a causal attention-to-sensitivity link ( $+8.5\% / -8.0\%$  from biasing attention toward/away from a change). **(5) The cueing effect is memory-driven rather than attention-driven, across an entire matched family:** in the multiplicative-gated model the cue and its validity are held in recurrent memory (100% decodable) while the attention map stays change-locked, and a matched family of six attention/memory wirings (multiplicative, dual-head, two-LSTM, cross-attention, dual-memory, and a failed affine modulation) confirms that *no* working variant orients attention to the cue on the short task — two of them attend markedly *away* from the cued image location. The same sweep names the network to carry forward: the two-LSTM multiplicative form is the best detector. We close with that recommendation for the empirical manuscript, a proposed task battery (VDA, set-size, Luo & Maunsell, Krauzlis), and a staged research plan.

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# Chapter 1

## Introduction

### 1.1 The program and its goal

The long-run objective of this program is a single neural-network architecture that (i) learns covert visual attention end-to-end from sparse, animal-like reward, (ii) reproduces the behavioural and neural signatures of primate spatial attention, and (iii) *generalises* across the standard battery of non-human-primate (NHP) attention tasks rather than being tuned to one. We are not merely fitting a curve; we are searching for the most neuro-inspired wiring that makes attentive behaviour *emerge* from the combination of a learnable priority computation, a capacity-limited working memory, and a reward signal — and that, having emerged, exposes an interpretable and causally manipulable internal attention signal.

The published manuscript this program supports, “*A recurrent vision transformer shows signatures of primate visual attention*” (Morgan, Albanna & Herman), establishes the empirical claim on the canonical four-stimulus, validity-driven, uniform-reward task. This report is the architecture study that sits behind that manuscript: it records what we tried, what worked, what failed, and — most importantly — *why*, so that the choice of network to carry forward is grounded in evidence rather than habit.

### 1.2 The model in one paragraph

The recurrent vision transformer parses each video frame into a small set of spatial patch tokens, runs a *recurrent* self-attention block in which the attention is computed jointly from the current image and the contents of a per-patch working memory, writes that memory with a spatial long short-term memory (LSTM) cell, and reads the memory with a distributional actor–critic head trained by reinforcement learning. The defining property is the recurrence of attention: because the attention at frame  $t$  depends on memory accumulated from previous frames, a cue shown briefly and then removed can be maintained and used to bias later processing. Every architectural decision in this report is, at bottom, a decision about *how* the image and the memory combine to form attention, and *which* internal state the decision is read from.

### 1.3 Scope of the evidence

The findings synthesised here rest on roughly thirty network variants trained on two task horizons, plus a clean-room reproduction of the exact published network. The variants span memory-as-tokens self-attention, cross-attention over memory, multiplicative (FiLM-style) gain modulation, vector-quantised codebook attention, predictive-coding and energy-modulation front-ends, split actor/critic pathways with and without cross-talk, fast-weight (hypernetwork) attention, and several novel dual-memory wirings. Two front-ends (raw-pixel and a reconstruction-

pretrained VAE) and two reinforcement-learning horizons (a seven-step fixed-onset task and a twenty-nine-step random-onset task) complete the design space. The reinforcement-learning machinery — prioritised episode replay, a maximum-a-posteriori-policy-optimisation actor loss, and a distributional quantile critic — is held essentially constant so that architectural comparisons are clean.

## 1.4 Contributions and structure

The contributions, each developed in its own chapter, are:

1. **Perception is the bottleneck on the short task** (Chapter 5). A reconstruction-pretrained VAE front-end is necessary and nearly sufficient to make the exact paper network learn the seven-step task; a front-end trained from reward alone never learns to represent orientation and the agent collapses to pressing at the deadline.
2. **The horizon decides which strategy is viable** (Chapter 6). The long task tolerates a weak front-end because the change can be integrated over many frames; the short task does not, which is why the VAE matters on one and not the other.
3. **Coupling is a precondition for learning** (Chapter 7). Severing the shared memory between the policy and value streams collapses an otherwise identical model to chance — shared memory is load-bearing, not cosmetic.
4. **The working network reproduces the paper’s behaviour** (Chapter 9): the reliability-scaled psychometric cueing benefit and the causal attention-to-sensitivity link both replicate cleanly.
5. **Cueing can be carried by memory rather than attention** (Chapter 10): a mechanistic dissociation that reframes how we read the attention maps and motivates a family of follow-up wirings.

The remaining chapters catalogue the full architecture zoo (Chapter 8), survey the reinforcement-learning harness and its variations (Chapters 4 and 12), and close with an architecture recommendation (Chapter 13), a proposed task battery (Chapter 14), a research plan (Chapter 15), and a one-page summary of the actionable findings (Chapter 16).

## 1.5 A note on terminology

Throughout, we call the two recurrent streams of the split-pathway models the *priority* ( $\mu$ ) pathway, which feeds the actor and the decision of *when* to declare a change, and the *value* ( $q$ ) pathway, which feeds the critic. We avoid the labels “salience” and “top-down” because the priority/commitment areas of the brain (LIP, FEF, SC) are each a single priority map rather than an actor/critic pair; the  $\mu/q$  split is a computational decomposition, and we are careful in Chapter 13 about the limits of mapping it onto anatomy.

## Chapter 2

# The task and its environments

### 2.1 The canonical cued change-detection task

The base environment is a spatially cued orientation-change-detection task built to reproduce the essential structure of primate covert-attention experiments. Each trial is a short video of  $50 \times 50$  grayscale frames. The canonical (paper) timeline is **seven frames**:

- $t = 0$ : black screen.
- $t = 1$ : a spatial *cue* appears at one stimulus location — top-left ( $S_1$ ) or bottom-right ( $S_4$ ). The cue is a disc whose surrounding ring is completed in proportion to the cue’s *validity* (the probability that, on a change trial, the change will occur at the cued location): 25%, 50%, 75%, or 100%.
- $t = 2$ : black screen (the cue is removed; its information must now be held in memory).
- $t = 3, 4$ : four Gabor-like stimuli appear, one per quadrant ( $S_1$  top-left,  $S_2$  bottom-left,  $S_3$  top-right,  $S_4$  bottom-right), at random orientations at least  $45^\circ$  apart, with per-frame orientation “wobble”.
- $t = 5$ : on half of trials (*change trials*) one stimulus’s orientation steps by  $\Delta$  degrees; on the other half nothing changes. The change persists to  $t = 6$ .
- $t = 6$ : final frame.

At every frame the agent chooses *wait* or *declare change*. Declaring ends the trial. A correct declaration at  $t \geq 5$  earns reward 1; waiting out a no-change trial to  $t = 6$  also earns 1; everything else earns 0. The two errors are therefore misses (failing to declare a real change) and false alarms (declaring on a no-change trial or declaring prematurely).

### 2.2 Difficulty control

Difficulty is set by two knobs. **Orientation noise**  $\sigma$  adds frame-to-frame jitter  $\delta_{it} \sim \mathcal{N}(0, \sigma)$  to each stimulus orientation;  $\sigma = 5$  in the canonical task. The **change magnitude**  $\Delta$  is drawn per change trial from  $\mathcal{U}(-k, k)$ , with  $k$  starting at  $65^\circ$  and shrinking by a curriculum (factor 0.9 each time rolling accuracy exceeds 0.85, floor  $8^\circ$ ) so the task hardens as the agent improves. For psychometric analysis we override  $\Delta$  to fixed magnitudes and freeze the curriculum, exactly as one would titrate difficulty in an animal.



## 2.3 Why the seven-step task is hard for reinforcement learning

The short horizon makes the task deceptively difficult to *learn* even though it is easy to *solve* (a scripted “wait until the change is visible, then press” policy scores 1.000). Two degenerate strategies sit at 0.5 accuracy and act as attractors: *always-wait* (correct on every no-change trial, misses every change) and *always-press-at-the-deadline* (catches every change, false-alarms every no-change). Because the change persists for only two frames, the reward-bearing “correct press” window is narrow, so the discriminating policy is hard to discover by exploration, and the policy entropy collapses before discovery. This is the central learning obstacle the perception and architecture findings of later chapters are really about.

## 2.4 The task battery

The same environment generalises to the standard NHP battery by editing the reward branch and the cue, which is how James proposes to broaden the empirical paper’s scope. The implemented and planned variants are:

- **validity4** — the canonical four-stimulus, uniform-reward, validity-driven task above. The workhorse of this report.
- **vda4** — value-cue variant: the cue colour (red/green/blue) sets the reward magnitude (5/3/1), so behaviour can be read for value direction as well as spatial cueing.
- **setsize{2,4,9}** — set-size manipulation; the nine-stimulus ( $3 \times 3$ ) version with five validity levels (including an uninformative 0) tests whether the cueing benefit *grows* with set size, the bridge to the normative set-size/accidental-hit theory (Paper B).
- **distractor** (the v7/v11\_part5 env) — the cue selects a *side*; the uncued side independently changes with probability 0.5 and must be ignored (pressing on it is a false alarm). Tests genuine selection against a competitor.
- **luo\_maunsell** — reward-structure manipulation (sensitivity vs. criterion sessions) after Luo & Maunsell (2018).
- **krauzlis** — attend/ignore manipulation after Krauzlis and colleagues, where an uncued change must be actively ignored.

All variants share the clean `ChangeDetectionEnv` interface: `reset` draws the cue, validity, and change; `step` exposes the two actions and the reward branch. This uniformity is what makes “train one architecture across the whole battery” a tractable research goal rather than a per-task engineering effort.

## 2.5 The long-horizon variant

A twenty-nine-step variant ( $T=29$ , change onset drawn uniformly in  $[11, 25]$ ) is used throughout the architecture history. It is the same task with a much longer post-change window, and as Chapter 6 shows, that single difference — the length of the window over which a change can be integrated — explains most of the otherwise puzzling discrepancies between the two horizons.

## Chapter 3

# The model architecture

The recurrent vision transformer has four stages: a patch front-end, a recurrent self-attention block, a spatial recurrent memory, and an actor–critic readout. This chapter gives the canonical (paper) form of each; later chapters vary one stage at a time.

### 3.1 Patch front-end

Each  $50 \times 50$  frame is divided into four  $25 \times 25$  quadrants. In the canonical network each quadrant is encoded by the encoder of a variational autoencoder (VAE) up to its second flattened layer, a 128-dimensional feature  $\mathbf{o}_{\text{flat},2}$  (the paper feeds the ViT this layer, not the stochastic latent). Per-patch positional and per-frame temporal one-hot codes are concatenated, giving a 140-dimensional token, and the four tokens form  $\mathbf{X}^{(t)} \in \mathbb{R}^{4 \times 140}$ . The role of the VAE — and the consequence of training it from reward versus from reconstruction — is the subject of Chapter 5.

### 3.2 Recurrent multiplicative self-attention

The canonical attention is single-head and *multiplicatively gated* by memory. With the previous-frame memory  $\mathbf{H}_1^{(t-1)} \in \mathbb{R}^{4 \times d_{\text{mem}}}$  ( $d_{\text{mem}} = 1024$ ),

$$\mathbf{Q} = (\mathbf{X} \mathbf{W}_{XQ}) \odot (\mathbf{H}_1 \mathbf{W}_{HQ}), \quad \mathbf{K} = (\mathbf{X} \mathbf{W}_{XK}) \odot (\mathbf{H}_1 \mathbf{W}_{HK}), \quad \mathbf{V} = (\mathbf{X} \mathbf{W}_{XV}) \odot (\mathbf{H}_1 \mathbf{W}_{HV}), \quad (3.1)$$

with  $\mathbf{W}_X \in \mathbb{R}^{140 \times 140}$ ,  $\mathbf{W}_H \in \mathbb{R}^{1024 \times 140}$ ,  $\odot$  the Hadamard product, and the attention

$$\mathbf{V}_{\text{flt}} = \text{softmax}(\mathbf{Q}\mathbf{K}^\top)\mathbf{V}, \quad \mathbf{Z}^{(t)} = \mathbf{X}^{(t)} + \mathbf{V}_{\text{flt}} \in \mathbb{R}^{4 \times 140}. \quad (3.2)$$

The multiplicative form is the program’s neuro-motivated gain-modulation: memory does not add to the image, it *scales* the bottom-up query/key/value projections. As Chapter 10 documents, this exact form has an important consequence — when memory is uninformative the gate is near zero and the attention is uniform — which is at the heart of the attention/memory dissociation. The cross-attention and FiLM variants replace Eq. (3.1) with, respectively,  $\mathbf{K}, \mathbf{V}$  formed by concatenating image and memory tokens, and a  $(1 + \mathbf{H}_1 \mathbf{W}_H)$  identity-floored gate.

### 3.3 Spatial recurrent memory

The percept  $\mathbf{Z}$  writes a per-patch memory through an exponential-gated spatial LSTM (an adaptation of the xLSTM), maintained independently per patch so that self-attention is the only route by which spatial locations exchange information:

$$\begin{aligned} \tilde{\mathbf{I}} &= \mathbf{Z}\mathbf{W}_i + \mathbf{H}\mathbf{R}_i, & \tilde{\mathbf{F}} &= \mathbf{Z}\mathbf{W}_f + \mathbf{H}\mathbf{R}_f, & \tilde{\mathbf{O}} &= \mathbf{Z}\mathbf{W}_o + \mathbf{H}\mathbf{R}_o, & \tilde{\mathbf{U}} &= \mathbf{Z}\mathbf{W}_u + \mathbf{H}\mathbf{R}_z, \\ \mathbf{M} &= \max(\tilde{\mathbf{F}} + \mathbf{M}', \tilde{\mathbf{I}}), & \mathbf{I} &= e^{\tilde{\mathbf{I}} - \mathbf{M}}, & \mathbf{F} &= e^{\tilde{\mathbf{F}} + \mathbf{M}' - \mathbf{M}}, & \mathbf{O} &= \sigma(\tilde{\mathbf{O}}), \end{aligned}$$

**Table 3.1.** Components of the canonical recurrent vision transformer and the axes along which the architecture zoo varies them.

| Stage           | Canonical form                                                         | Variation axis explored                                                   |
|-----------------|------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Front-end       | VAE encoder $\rightarrow \mathbf{o}_{\text{flat},2}$ (128-d), 4 tokens | raw-pixel vs. conv vs. frozen VAE; 4 vs. 100 tokens                       |
| Attention       | multiplicative gate, single head                                       | cross-attention, FiLM (1+.), fast-weights, codebook, dual-head            |
| Feedback source | memory $\mathbf{H}_1$ scales Q/K/V                                     | memory as keys/values; memory generates weights; memory as queries        |
| Memory          | one xLSTM, $d_{\text{mem}}=1024$                                       | two LSTMs (feedback vs. decision); parallel vs. stacked; LSTM vs. vanilla |
| Readout         | flattened $\mathbf{H}$ (4096)                                          | read $\mathbf{Z}$ ; split $\mu/q$ ; codebook selection                    |

$$\mathbf{N} = \mathbf{F} \odot \mathbf{N}' + \mathbf{I}, \quad \mathbf{C} = \mathbf{C}' \odot \mathbf{F} + \tanh(\tilde{\mathbf{U}}) \odot \mathbf{I}, \quad \mathbf{H} = \mathbf{O} \odot (\mathbf{C}/\mathbf{N}),$$

with  $d_{\text{mem}} = 1024$ ,  $\mathbf{W}_x \in \mathbb{R}^{140 \times 1024}$ ,  $\mathbf{R}_x \in \mathbb{R}^{1024 \times 1024}$ , and the stabiliser/normaliser states  $\mathbf{M}, \mathbf{N}$ . The memory is the agent’s working memory: it carries the cue and the accumulating evidence across the blank interval.

### 3.4 Actor–critic readout

The four-patch structure is retained through attention and memory but *flattened* at the reinforcement-learning agent: the readout is  $\mathbf{H}' \in \mathbb{R}^{4096}$ . A four-layer feed-forward actor maps  $\mathbf{H}'$  to a two-way policy (wait / declare); a distributional critic maps  $\mathbf{H}'$  to a value distribution. Which internal state is read — the flattened memory  $\mathbf{H}$  (canonical) versus the attention output  $\mathbf{Z}$  (the cross-talk and codebook variants) — is itself an architectural variable that, we find, changes the phenotype.

## Chapter 4

# The reinforcement-learning harness

The agent learns purely from the sparse task reward; there is no reconstruction or auxiliary loss on the recurrent network (the VAE, when used, is pretrained separately and frozen). The harness has been held essentially constant across architectures so that comparisons are clean, and it is itself one of the program’s hard-won results: a stable recipe for training a bootstrapped recurrent actor–critic on a task with a deceptive 0.5 accuracy floor.

### 4.1 Actor: PAC (maximum-a-posteriori policy optimisation + behaviour cloning)

The actor loss replaces the clipped PPO surrogate with the PAC objective. For the discrete action set the maximum-a-posteriori-policy-optimisation E-step is closed form,

$$q(a \mid s) = \text{softmax}_a[\log \pi_{\theta'}(a \mid s) + \bar{Q}_{\theta'}(s, a)/\eta], \quad (4.1)$$

where  $\theta'$  is a time-lagged target network,  $\bar{Q}$  is the critic’s mean over the value distribution, and  $\eta$  is a temperature. The M-step plus a behaviour-cloning term gives

$$\mathcal{L}_{\text{actor}} = (1 - \alpha) \left( - \sum_a q(a \mid s) \log \pi_{\theta}(a \mid s) \right) + \alpha \left( - \log \pi_{\theta}(a_t \mid s) \right),$$

with  $\eta = 0.1$ ,  $\alpha = 0.1$ . The critic’s value drives the actor *directly*; there is no generalised-advantage estimation.

### 4.2 Critic: distributional quantile regression

The critic is a distributional QR-DQN critic: it predicts a set of value quantiles and is trained with the quantile-Huber loss against the one-step distributional temporal-difference target  $r_t + \gamma \mathbf{V}_{\text{dist}}(s_{t+1})(1 - d_t)$ . The published network used a 15-bin categorical critic; in the reproduction we use 5–15 quantiles. The number of “particles” is a deliberate knob (Chapter 12).

### 4.3 Prioritised episode replay

Fresh on-policy episodes are augmented with replay episodes drawn at episode granularity  $\propto \text{priority}^\alpha$ , priority being the mean per-step quantile-Huber error, with importance-sampling weights annealed over training. Episode-level replay is essential on this task: the discriminating “correct press” events are rare, and replay keeps them in the gradient long enough to be learned.

## 4.4 Target network: hard-copy versus EMA

The target network  $\theta'$  supplies both the critic’s bootstrap target and the actor’s E-step reference policy. Two update rules are available: a periodic hard copy ( $\theta' \leftarrow \theta$  every  $N$  optimiser steps, the form that produced the original working checkpoints) and an exponential moving average ( $\theta' \leftarrow d\theta' + (1-d)\theta$  each iteration). The EMA form was added during the reproduction; whether it is load-bearing is one of the open A/B questions of Chapter 12.

## 4.5 Burn-in

For the first  $\sim 20$  iterations the agent is forced to wait, training only the critic on full-length episodes so the value function and recurrent representation warm up on clean trajectories before the policy acts. This is the descendant of the program’s earlier reconstruction-pretrain phase and materially reduces the early collapse to premature pressing.

## 4.6 What the harness taught us

Three lessons recur in the history. First, **always-wait is a stable harbor**: a bootstrapped actor on this task will retreat to the safe action that never incurs a false-alarm penalty unless exploration is actively maintained. Second, **the target network is the main stabiliser**: a time-lagged reference for both the critic target and the E-step reference policy is what keeps the bootstrap from chasing its own tail. Third, **a degenerate 0.5 optimum has two faces** (always-wait and always-press-at-deadline) and which one a run falls into is seed-dependent — both are collapses, and escaping them is a perception/exploration problem, not a critic problem (the critic learns the 0.5 value perfectly well in both).

## Chapter 5

# Finding I — perception is the bottleneck

The single most consequential result of the reproduction is that, on the seven-step task, the obstacle to learning is not the reinforcement-learning objective, the attention mechanism, or the memory — it is *perception*. A front-end trained from reward alone never learns to represent orientation, and the agent therefore cannot see the change it is asked to report.

### 5.1 The collapse

The exact published network, trained end-to-end for the full 150,000 episodes with the VAE encoder learned from reward, does not reproduce the task. It converges to *always-press-at-the-deadline*: it presses at frame 5 (the first frame a change could appear) on every trial, scoring hit rate 1.000 on change trials and false-alarm rate 1.000 on no-change trials, for an overall accuracy of 0.515 and a reaction time pinned at exactly five frames. It has learned the *timing* of the change window but not the *discrimination* of change from no-change.

### 5.2 The diagnosis

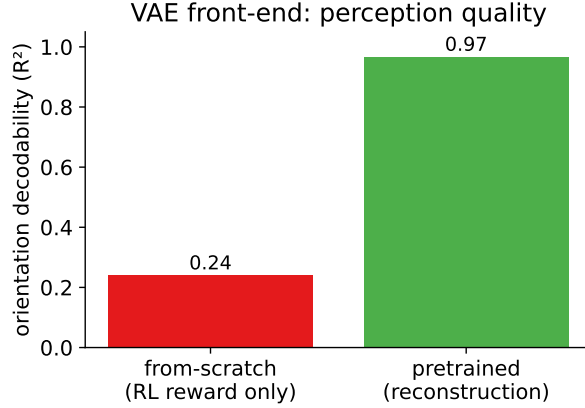
Why can it not discriminate? Because to detect an orientation change one must represent orientation, and the from-scratch front-end does not. We probed this directly: render single Gabor patches at known orientations, encode them through the trained front-end’s  $\mathbf{o}_{\text{flat},2}$  layer, and fit a linear decoder for the (circular) orientation. The result is decisive (Fig. 5.1):

$$R^2_{\text{from-scratch}} = 0.24, \quad R^2_{\text{pretrained}} = 0.97.$$

The reward-trained front-end’s features barely encode orientation; a reconstruction- pretrained VAE’s features encode it almost perfectly. Sparse reward (on the order of one bit per episode) simply cannot teach fine orientation perception, which is exactly why the paper pretrains the VAE as a feed-forward preprocessing step — a step our first reproduction had omitted.

### 5.3 The rescue

Pretraining the VAE is cheap and decisive. We sample  $\sim 42,000$  grayscale  $25 \times 25$  patches from the environment (blanks, cues, Gabors), train the paper’s exact VAE (convolutional encoder  $\rightarrow \mathbf{o}_{\text{flat},2} \rightarrow \text{latent}$ , transposed-convolution decoder, reconstruction + small KL) for  $\sim 12$  epochs to a reconstruction loss of 0.0025, and freeze the encoder into the front-end. With this single change — *nothing else in the network or harness altered* — the exact paper network **learns the task on the first attempt** and climbs above the 0.5 floor quickly, reaching hit rate 0.79



**Figure 5.1.** Orientation decodability ( $R^2$  of a linear decoder on the front-end’s  $\mathbf{o}_{\text{flat},2}$  features) for the front-end trained from reward versus pretrained on reconstruction. Reward alone cannot teach orientation perception; reconstruction can.

with a false-alarm rate of 0.08 and a median reaction time of five frames (it presses the instant it perceives the change).

## 5.4 The VAE is a static, task-parameter-invariant front-end

A subtle but useful corollary: the VAE is a *per-frame, static* encoder. It reconstructs one patch at a time and has no notion of motion or time. Task temporal parameters — the orientation-noise  $\sigma$  and the horizon  $T$  — do not change what the VAE sees, because the displayed orientations span the full range regardless of  $\sigma$ . A single pretrained VAE therefore serves all noise levels and both horizons; there is no need to re-pretrain per task variant. This both simplifies the pipeline and sharpens the interpretation of the next chapter: when the long task succeeds without the pretrained VAE, it is not because the perception problem went away, but because the long horizon admits a different solution to it.

## 5.5 Why this matters for the architecture choice

This finding cleanly separates two questions that the earlier history had conflated. “Does the architecture work?” and “Can the front-end see?” are different. Many of the collapses catalogued in Chapter 8 on the seven-step task are, in retrospect, perception failures wearing an architecture’s clothes. The methodological upgrade is to *fix perception first* (frozen pretrained VAE) and only then compare attention/memory wirings — which is precisely the controlled setting of Chapters 9 and 10.

A third perception result (Chapter 11) qualifies this: perception need not be solved *before* the architecture if the network carries a self-supervised predictive auxiliary — a V-JEPA temporal self-distillation loss — trained jointly with the policy, which learns a working front-end from raw pixels or a convolutional encoder with no pretrained VAE.

## Chapter 6

# Finding II — the horizon decides the strategy

### 6.1 The puzzle

The history contains a genuine paradox. The cross-talk network reproduced primate cueing behaviour on the *long* twenty-nine-step task (hit  $\approx 0.80$ ) trained from reward with no pretrained VAE. The *same* architecture, and the exact paper network, collapse on the *short* seven-step task without a pretrained VAE — even though the short task is, by every naive measure, easier (fewer frames, less accumulated noise, a fixed change time the agent could memorise). How can the easier task be the one that fails?

### 6.2 Two ways to detect a change

The resolution is that there are two strategies for change detection, and they have opposite horizon dependence.

**Instantaneous detection.** Compare *this* frame’s orientation to the remembered baseline.

This needs accurate per-frame orientation perception, and it detects the change in a single frame.

**Temporal / flow detection.** Notice that *something shifted* in the visual stream across frames, without representing what the orientation is. The per-frame signal is weak and noisy, so it must be integrated over many frames to become reliable; it is slow.

### 6.3 The horizon selects between them

On the long task the change persists for up to  $\sim 18$  frames, so the flow strategy works: the agent accumulates a weak “something moved” signal until it is confident. It never *needs* accurate perception, and consequently never learns it — which is exactly why the long-trained front-end has orientation decodability  $R^2 \approx 0.24$  yet still solves the long task. On the short task the change lasts two frames. There is no time to integrate, the flow strategy fails, and the only thing that works is instantaneous perception — which sparse reward cannot build (Chapter 5). The pretrained VAE supplies exactly the missing instantaneous perception, and the short task is solved.

In one sentence: **the short task removes the temporal-integration escape hatch and forces abstract, single-frame perception.** Pixel/flow inference is slow; abstract-but-accurate inference is fast.



## 6.4 The behavioural fingerprint

The theory makes a sharp, falsifiable prediction that the data already bear out. A model that detects instantaneously should press the moment the change appears, so its reaction time should be pinned at change onset and *cannot* be sped up by the cue. The pretrained-VAE model on the short task does exactly this: median reaction time is five frames (change onset), essentially flat across change magnitude and across cue validity (Chapter 9). A flow-based model, by contrast, should have long, magnitude- dependent reaction times. The prediction to complete the argument is a direct comparison of reaction-time distributions: the from-scratch long-task model should press *late* and the pretrained-VAE model *early*, the clean signature of “flow is slow, perception is fast.”

## 6.5 Consequences for the program

Two consequences follow. First, **the short task is the more discriminating test bed**: because it forecloses the flow shortcut, it forces the network to actually perceive, and so it exposes architectural differences that the long task hides behind temporal integration. This is why the cue-attention investigation of Chapter 10 is run on the short task. Second, **horizon is a design variable, not a nuisance**: choosing the seven-step task with the frozen VAE gives the cleanest, most mechanism- revealing setting, while the twenty-nine-step task is the right venue for studying the *maintenance* of attention across a delay. The empirical paper will want both: the short task for the sensitivity/criterion story and the long task for the maintenance and anticipation story.

## Chapter 7

# Finding III — coupling is a precondition for learning

A second structural result, established on the long task with the `v11` family of split-pathway networks, is that the coupling between the policy and value streams is not an optimisation detail — it is a precondition for the task to be learned at all.

### 7.1 The split-pathway design

The `v11_part2` network splits the recurrent encoder into two single-head cross-attention streams that share a deep memory. The *priority* ( $\mu$ ) stream queries the image and attends over  $[\mathbf{H}_1 \parallel \mathbf{H}_2]$  with an image residual, and its output  $\mathbf{Z}_\mu$  drives the actor. The *value* ( $q$ ) stream queries the image and attends over  $\mathbf{H}_2$  with a memory residual, and its output  $\mathbf{Z}_q$  drives the critic. The cross-talk lives in the shared deep memory  $\mathbf{H}_2$ : the priority stream *writes* it (through  $\mathbf{Z}_\mu$ ) and *reads* it back, while the value stream reads the same  $\mathbf{H}_2$ . The two heads are separated but sit on a jointly maintained recurrent register.

### 7.2 The three-way comparison

Holding the front-end, heads, parameter count, and entire training recipe fixed, three wirings were trained and evaluated (Table 7.1):

1. **Shared memory (cross-talk).** As above: the two streams coupled through  $\mathbf{H}_2$ , priority driving the actor.
2. **Independent (no cross-talk).** The same two stream *types*, but each with its own private memory updated only by its own output — the minimal edit that removes the coupling while preserving the split.
3. **Swapped routing.** Architecturally identical to the coupled model, but the value stream drives the actor and the priority stream drives the critic.

### 7.3 The result

The coupled model is a genuine psychophysical detector. Both uncoupled controls collapse completely: across hundreds of episodes they never once declare a change. The independent model’s failure shows that **a shared substrate is necessary** — decoupling the critic’s state from the actor’s state breaks credit assignment, because the value estimate that should teach the policy is computed on memory the policy cannot influence. The swapped-routing failure shows

**Table 7.1.** Coupling between the policy and value pathways is necessary for learning. Same front-end, heads, parameter count and training recipe; only the inter-pathway wiring differs. Numbers from the trained checkpoints on the long task.

| Configuration               | Accuracy     | Hit rate     | FA rate | Phenotype                   |
|-----------------------------|--------------|--------------|---------|-----------------------------|
| Shared memory (cross-talk)  | <b>0.868</b> | <b>0.794</b> | 0.067   | active, calibrated detector |
| Independent (no cross-talk) | 0.504        | 0.000        | 0.000   | always-wait collapse        |
| Swapped routing             | 0.420        | 0.000        | 0.000   | always-wait collapse        |

that **which stream drives the policy also matters** — the grounded, image-reading priority stream must command the actor; routing the memory- grounded value stream to the policy fails even with the coupling intact. The two conditions are jointly necessary, and the margin — a fifty-point accuracy gap from a single wiring edit — is among the largest the program has produced.

## 7.4 Interpretation and its limits

The result echoes a deep feature of the biology: in a cortico-basal-ganglia actor-critic, the actor is updated only through a dopaminergic teaching signal gated by the critic’s reward-prediction error, so severing value from action breaks learning. We are careful, however, about the mapping. The brain’s coupling is a largely scalar, diffusely broadcast critic-to-actor signal, whereas the model’s coupling is a shared recurrent working-memory register that the actor writes and both read. The model therefore *formalises one concrete way* to satisfy a constraint the brain satisfies by other means, rather than replicating the mechanism. The asymmetry result — that the grounded stream must drive the policy — is better motivated by the evidence-accumulation literature (the sensory stage is the causally necessary one) than by the actor-critic literature, and is the sharpest, most specific claim the split-pathway work makes.

## 7.5 Standing caveat

The cross-talk results were obtained on the long task, before the perception finding of Chapter 5. The split-pathway models read **Z** (the attention output), not **H** (memory), which on the short task interacts with the perception and attention findings in ways Chapter 10 takes up directly. The coupling result itself is robust; its expression on the short task with a frozen VAE is one of the open items in the research plan.

## Chapter 8

# The architecture zoo

This chapter catalogues the network variants the program has built and, where trained, their outcomes. The point is not encyclopaedic completeness but the *pattern* of what makes attentive behaviour emerge versus collapse. Table 8.1 is the index; the prose groups the variants into the families that carry the lessons.

### 8.1 Family A — memory-as-tokens self-attention (v4, v5, v5\_part2)

The earliest mature line treats the recurrent states as additional tokens. v5 self-attends over the concatenation  $[\text{patch} \parallel \mathbf{H}_1 \parallel \mathbf{H}_2]$  at every layer. It works and is highly interpretable: the encoder’s first layer becomes a *top-down, cue-driven spatial orienter* (heads spike to the cued quadrant at cue onset, follow the cued side, and scale with cue reliability — the substrate of the validity effect), while the actor/critic decoder’s final layer is a *bottom-up change detector* that snaps onto the changed quadrant at change onset. This encoder–decoder dissociation (top-down cue prior vs. bottom-up change read-out) is a recurring motif. v5\_part2 re-architects this as eight-head cross-attention over memory followed by stacked per-token LSTMs and one-dimensional-convolution heads; it works, and the load-bearing computation moves into the encoder’s memory cross-attention (memory keys are causally decision-moving, image keys are inert). v4 (a FiLM-encoder sibling) revealed a cautionary failure mode: a FiLM gain that initialises to zero with no “+1” offset makes the attention a structurally uniform global average-pool — the first appearance of the “identity floor matters” lesson that returns in Chapter 10.

### 8.2 Family B — bottleneck residuals (v8, v8\_part2, v9)

This line tightens how visual information reaches memory. v8 routes the attention residual through the previous memory rather than the image, so visual content can only flow through attended values; it works, with very fast reaction times and an emergent dedicated patch-reading head. Push the bottleneck further — memory-only keys and values (v8\_part2) or queries drawn from memory rather than the image (v9) — and the models collapse to always-wait. The diagnosis is consistent: removing the grounded image fallback puts perception on a knife-edge, so the policy retreats to the safe action. This is the same lesson the split stream of v11 was designed to fix.

### 8.3 Family C — split priority/value pathways (v11, v11\_part2-5)

The split-pathway family is the program’s most productive line and the subject of Chapter 7. v11 (parallel streams, heads read both transformer outputs) fixed the always-wait collapse by grounding the priority stream in the image. v11\_part2 added the deep memory to the priority

keys and split the readout (priority  $\rightarrow$  actor, value  $\rightarrow$  critic); it is the canonical working cross-talk model, and its headline is the *reliability-scaled* cueing benefit that the un-split **v11** lacked. **v11\_part3** (swapped readout) and **v11\_part4** (independent, no cross-talk) both collapse to chance — the necessity results of Table 7.1. **v11\_part5** ports the working cross-talk model to the distractor task and yields a clean double dissociation: biasing the priority stream swings the decision while leaving value flat, biasing the value stream moves value but never the decision; the distractor is filtered in the *decision* pathway (decodable from sensory memory but cold to the priority map — “seen but not acted upon”), and value gates *whether* the priority filter is deployed in a binary, engage/abandon fashion.

## 8.4 Family D — codebook / vector-quantised attention (**v12**, **v12\_part2/3**)

Motivated by an “attention as a lookup table / superior-colliculus microstimulation” analogy, this line makes the attention values a learnable codebook. The hard sampled/STE version failed to learn (a gradient mismatch under prioritised replay re-encoding); the soft-attention codebook (**v12**) is differentiable and trains. The adaptive-codebook extensions add a deformation with an energetic maintenance cost (**v12\_part2**) and an EMA codebook written by a gated image-content stream (**v12\_part3**); the EMA-only variant collapses (a catch-22 in which the magnitude penalty drives the very signal the EMA should consolidate to zero), which is what motivated the gradient-learned-codebook fix.

## 8.5 Family E — predictive-coding and energy modulation (**v13**, **v14**, **v15**)

The most exploratory line asks whether a self-supervised recurrent dynamical system can be repurposed for the task by “energy modulation” rather than retraining. **v13** freezes a JEPA-style predictive-coding encoder and adds an energy stream that modulates its dynamics, training only the energy and heads. **v14** runs two gradient-firewalled modules concurrently — a self-supervised module (JEPA latent, no pixels) and a reinforcement-learning module — that share state forward but never gradients. **v15** adds a VQ-VAE-style energy term on top of a gradient-learned codebook that provably reduces to the working codebook when the energy is off. These are mechanism studies more than performance plays; on the current low-bandwidth task they are expected to track their working bases, with the energy/adaptation machinery reserved for higher-bandwidth tests.

## 8.6 What the zoo says in aggregate

Three regularities cut across the families. (1) **A grounded image pathway is required:** every variant that removes the bottom-up route to either memory or the decision collapses to always-wait. (2) **Routing, not representation, decides the phenotype:** identical encoders produce a working detector or a chance model depending only on which stream drives the policy and whether a shared substrate couples value to action. (3) **Interpretability is an emergent property of the right wiring:** the variants that learn also expose clean, manipulable attention maps (cue-orienting encoders, change-locked decoders), whereas the collapsed variants expose nothing. These regularities are the empirical backbone of the recommendation in Chapter 13.

**Table 8.1.** Index of the architecture zoo. “Outcome” is the trained behaviour where a run exists; many variants are mechanism studies built and unit-tested but reserved for a higher-bandwidth task. ✓ = learns/works, × = collapses, ○ = built, not yet a decisive run.

| Variant   | Mechanism (what is varied)                                                    | Outcome                     |
|-----------|-------------------------------------------------------------------------------|-----------------------------|
| v5        | memory-as-tokens self-attention, [patch    $\mathbf{H}_1$    $\mathbf{H}_2$ ] | ✓ (cue-orienting)           |
| v5_part2  | 8-head cross-attn over memory → stacked LSTMs → conv heads                    | ✓                           |
| v4        | FiLM-encoder sibling; revealed zero-init gain ⇒ uniform attn                  | ○ (diagnostic)              |
| v7        | v5_part2 arch on cued-side distractor task                                    | ○                           |
| v8        | H1-residual bottleneck (visual info only via attended values)                 | ✓ (fast RT)                 |
| v8_part2  | memory-only K/V (patch = queries only)                                        | × (always-wait)             |
| v8_part3  | predictive-comparator write gate (surprise opens LSTM2 input)                 | ○                           |
| v9        | memory-query attention (queries from $\mathbf{H}_1$ , not image)              | × (collapse)                |
| v11       | parallel priority    value streams; heads read both $\mathbf{Z}$              | ✓ (hit 0.63)                |
| v11_part2 | + deep memory in priority keys, split readout (the cross-talk)                | ✓ (hit 0.80)                |
| v11_part3 | v11_part2 with actor/critic readout swapped                                   | × (chance)                  |
| v11_part4 | independent streams, no shared memory                                         | × (chance)                  |
| v11_part5 | cross-talk on the distractor task                                             | ✓ (double dis-soc.)         |
| v12       | soft codebook attention, static learnable codebook                            | ✓ (soft)                    |
| v12_part2 | adaptive codebook deformation + energetic cost                                | ○                           |
| v12_part3 | EMA codebook written by a gated content stream                                | × (EMA-only)<br>/ ○ (gated) |
| v13       | frozen predictive-coding encoder + energy-modulation stream                   | ○                           |
| v14       | two gradient-firewalled modules (self-supervised ⊥ RL)                        | ○                           |
| v15       | gradient codebook + VQ-VAE energy (reduces to v12)                            | ○                           |
| setsize9  | cross-talk arch on the 9-stimulus set-size task                               | ○                           |

## Chapter 9

# Finding IV — the working model reproduces the paper

With perception fixed by the frozen VAE (Chapter 5), the exact paper network on the seven-step task is a real psychophysical detector, and it reproduces the core behavioural figures of the published manuscript. This chapter is the quantitative deep dive on that model.

### 9.1 Baseline behaviour

Under a greedy policy the model catches 79% of real changes, false-alarms on 8% of no-change trials, and presses with a median reaction time of five frames — the instant a change can appear. It is not a degenerate 0.5 model; it discriminates change from no-change with a healthy sensitivity.

### 9.2 Psychometrics: the reliability-scaled cueing benefit

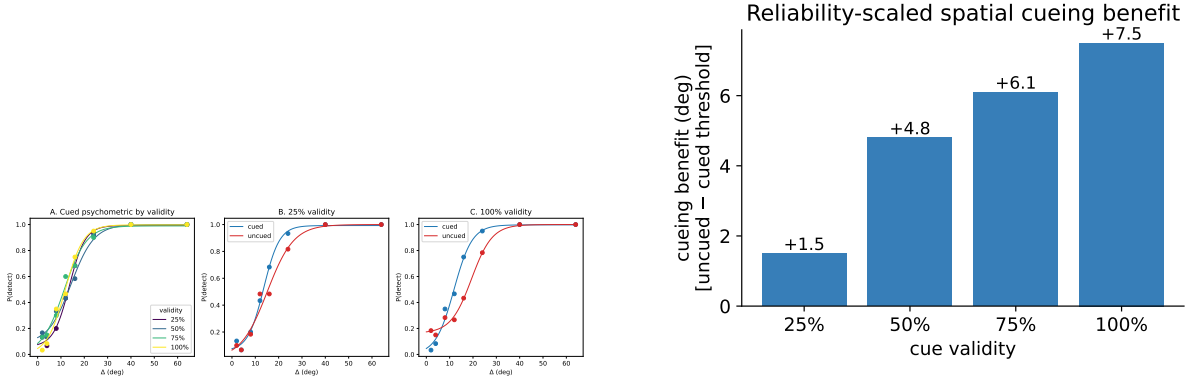
The discriminating test of attention is not accuracy but the *cueing signature*: a detection advantage for changes at the cued location that scales with how reliable the cue is. We measured it by presenting a cue of a given validity, forcing the change at the cued ( $S_1$ ) or uncued ( $S_4$ ) location at a fixed magnitude  $\Delta$ , and fitting a four-parameter logistic to the detection-rate-versus- $\Delta$  curve. The threshold (the logistic’s position) is the sensitivity readout. The result (Fig. 9.1, Table 9.1) is the paper’s headline: the cued threshold falls below the uncued threshold by an amount that *grows monotonically with validity* — a negligible  $+1.5^\circ$  at 25%, a robust  $+7.5^\circ$  at 100%. At 25% validity (no predictive spatial information) there is essentially no cueing effect; at 100% there is a large one, carried by the threshold/sensitivity parameter rather than the guess or lapse rates. This is the canonical sensitivity-enhancing signature of primate spatial attention, reproduced.

### 9.3 Chronometrics: the reaction-time story

Reaction time is pinned at change onset ( $\approx 5$  frames) with only a faint difficulty effect (slightly slower for the hardest changes) and *no* reaction-time cueing benefit (cued and uncued reaction times are equal to within 0.06 frames; Fig. 9.2). This is not a failure to reproduce so much as a consequence of the instantaneous-perception mechanism (Chapter 6): the cue moves the decision *criterion* (accuracy) but cannot speed a response that is already at the floor. It is a specific, testable prediction — good perception yields a criterion-shift cueing effect without a reaction-time benefit — and it is one of the places the model diverges, informatively, from a slower flow-based detector.

**Table 9.1.** Psychometric thresholds (four-parameter logistic position, degrees) for cued versus uncued changes by cue validity, and the resulting cueing benefit. The benefit scales with validity — the paper’s central behavioural result.

| Cue validity | Cued threshold | Uncued threshold | Cueing benefit |
|--------------|----------------|------------------|----------------|
| 25%          | 13.6           | 15.1             | +1.5           |
| 50%          | 14.3           | 19.2             | +4.8           |
| 75%          | 11.3           | 17.5             | +6.1           |
| 100%         | 11.8           | 19.4             | +7.5           |



**Figure 9.1.** Left: psychometric detection-rate curves by cue validity (cued changes) and the cued-versus-uncued comparison at low and high validity. Right: the cueing benefit (uncued–cued threshold) scales monotonically with validity.

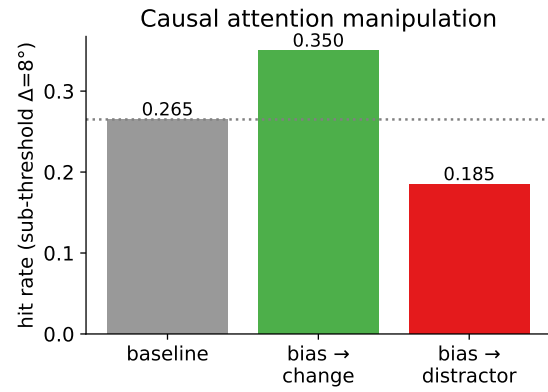
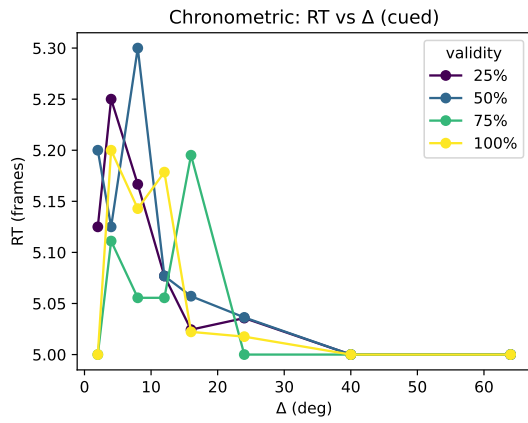
## 9.4 Causal attention manipulation

The paper’s causal result is that attention is causally linked to perceptual sensitivity, not merely correlated with it. We tested this by clamping the attention onto a target patch during the stimulus frames and measuring detection of a sub-threshold change (Fig. 9.2, right). Biasing attention *toward* the changed location improves the hit rate from 0.265 to 0.350 (+8.5%); biasing it *toward a distractor* impairs it to 0.185 (−8.0%); and biasing attention on a no-change trial barely moves the false-alarm rate (+1%). The effect is bidirectional and clean — exactly the hallmark causal signature of attention (enhanced sensitivity at the attended location, like sub-threshold superior-colliculus or frontal-eye-field microstimulation). Crucially, biasing attention does not *manufacture* a change percept; it gates *which* location’s content reaches the decision.

## 9.5 Scorecard

On the seven-step task the working model reproduces the two results the published manuscript is built on — the reliability-scaled psychometric cueing benefit and the causal attention-to-sensitivity link — cleanly. The reaction-time cueing benefit does not reproduce (reaction time is floored by instantaneous detection), and the attention- *dynamics* story is its own subject, taken up next.





**Figure 9.2.** Left: reaction time versus change magnitude by cue validity — pinned near five frames, a weak difficulty effect, no validity benefit. Right: the causal attention manipulation — biasing attention toward the changed location improves detection, biasing it toward a distractor impairs it, without inducing false alarms.

## Chapter 10

# Finding V — attention versus memory, and the cue-attention question

The deep dive raised a mechanistic question that is, at the time of writing, the program’s live frontier: in the canonical multiplicatively-gated model, the cueing *behaviour* is real and graded, yet the per-frame attention map does not orient to the cue. This chapter sets out the dissociation, the mechanism behind it, and the matched family of architectures we are training to resolve it.

### 10.1 The cue modulates behaviour

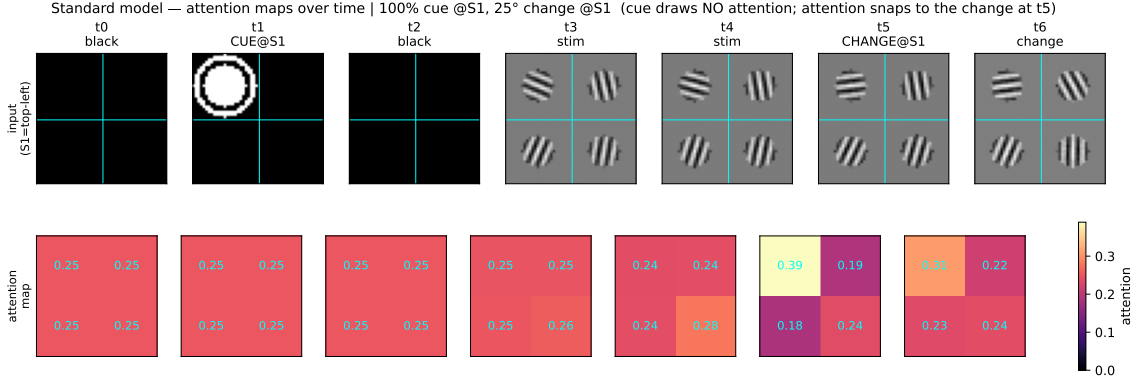
First, to be unambiguous: the cue strongly modulates the model’s behaviour. The detection threshold for a cued change is up to  $7.5^\circ$  lower than for an uncued change, and the benefit scales with validity (Chapter 9). The model reads the cue, holds it, and uses it. The question is not *whether* the cue is used but *through what internal variable*.

### 10.2 The attention is change-locked, not cue-locked

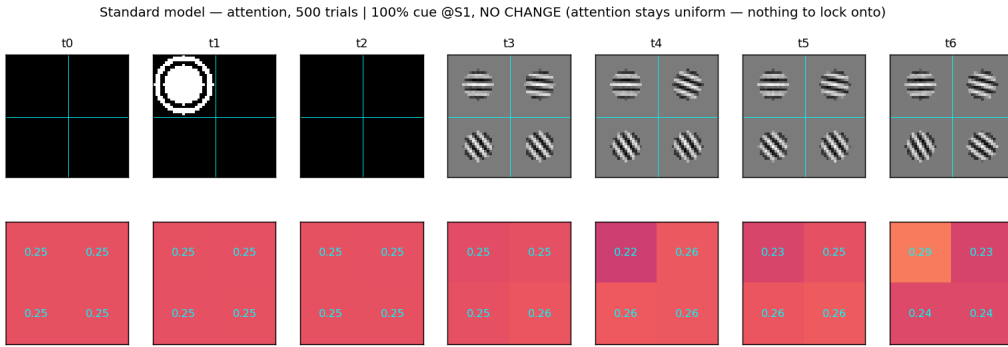
We measured the four-patch attention map at every frame for a 100% cue at  $S_1$  with a  $25^\circ$  change at  $S_1$  (Fig. 10.1). The attention is uniform ( $\approx 0.25$  per patch) through the cue frame and the entire delay, and it sharpens onto  $S_1$  only at frame 5 — the moment the change appears. The full  $4 \times 4$  query-key matrix at the cue frame is essentially uniform, so this is not an artefact of how the map is summarised: there is no cue-driven structure to summarise. A no-change control confirms the reading (Fig. 10.2): with the same 100% cue but no change, the attention stays flat across all seven frames (maximum deviation from uniform 0.039). The attention fires at a location when, and only when, that location *changes*.

### 10.3 Where the cue actually lives

If not in the attention, where? In the recurrent memory. Decoding from the flattened memory  $\mathbf{H}$  at the pre-change frame, both the cued *location* ( $S_1$  vs.  $S_4$ ) and the *validity* (25/50/75/100%) are recovered at 100% accuracy (chance 50% and 25%). The cue and its reliability are perfectly maintained in working memory, where they set the decision threshold; the attention, meanwhile, is a bottom-up change detector. This is a clean dissociation: **cue**  $\rightarrow$  **behaviour: yes**; **cue**  $\rightarrow$  **attention map: no**.



**Figure 10.1.** Per-frame attention (top: input frames; bottom:  $2 \times 2$  attention map, S<sub>1</sub> = top-left) for a 100% cue at S<sub>1</sub> with a 25° change at S<sub>1</sub>. The cue (frame 1) draws no attention; the attention concentrates on S<sub>1</sub> only at the change (frame 5).



**Figure 10.2.** No-change control, 500 trials, 100% cue at S<sub>1</sub>. With nothing to lock onto the attention stays uniform across every frame — the cue alone never moves it.

## 10.4 The mechanism

The dissociation has a transparent cause in the multiplicative form (Eq. 3.1). Because  $\mathbf{Q} = (\mathbf{X}\mathbf{W}_{XQ}) \odot (\mathbf{H}_1\mathbf{W}_{HQ})$ , the query magnitude is gated by the memory projection. At the cue frame the memory has not yet built up, the gate  $|\mathbf{H}_1\mathbf{W}_{HQ}|$  is near zero ( $\approx 0.02$ ), and the queries and keys collapse toward zero, so the attention is uniform. The model is then *free* to keep that gate small, because the cue reaches memory through the image residual  $\mathbf{X}$  in  $\mathbf{Z} = \mathbf{X} + \mathbf{V}_{\text{filt}}$  and the LSTM, without the attention ever needing to orient. The multiplicative form, in short, *admits* a memory-only solution, and on the seven-step task with a frozen VAE and a memory readout, the harness finds it.

## 10.5 The open question

This is where honesty is required. The lab’s other models — multi-head, cross-attention variants in the spirit of v5 and v11 — reliably *do* orient attention to the cue, across many trained networks. So orienting is clearly achievable with these mechanisms; the multiplicative-gated reproduction is the outlier. The cause is therefore *some feature of our specific reproduction* — the gate form, the head count, the memory readout, the harness, the frozen front-end — not a property of memory-gated attention in general. We do not yet know which feature, and we are not guessing: we are testing it directly with a matched family of architectures.

**Table 10.1.** The matched family of attention/memory wirings under training to resolve the cue-attention question. All share the frozen-VAE front-end, the PAC+QR-DQN+PER harness, and the 4096-d readout.

| Variant        | What it changes about the feedback / memory                                                                                                                         | Status                      |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| multiplicative | canonical $\mathbf{Q} = (\mathbf{X}\mathbf{W}) \odot (\mathbf{H}_1\mathbf{W})$ (the reproduction)                                                                   | works; change-locked        |
| affine         | memory derives a scale-matrix + shift: $\mathbf{X}' = \Gamma(\mathbf{H}_1)\mathbf{X} + \beta(\mathbf{H}_1)$                                                         | <b>failed</b> (always-wait) |
| dualhead       | a memory-free sensory head    the multiplicative feedback head                                                                                                      | works; weak orient          |
| two-lstm       | split memory: $\mathbf{H}_1$ feeds attention, $\mathbf{H}_2(\mathbf{H}_1)$ feeds heads                                                                              | <b>best detector</b>        |
| crossattn      | additive cross-talk: $\mathbf{K}, \mathbf{V} = [\mathbf{X} \parallel \mathbf{H}_1]$ , two LSTMs from $\mathbf{Z}$                                                   | works; memory-routed        |
| dualmem        | memories <i>query</i> : $\mathbf{Q}([\mathbf{H}_1 \parallel \mathbf{H}_2])$ , $\mathbf{K}, \mathbf{V} = [\mathbf{X} \parallel \mathbf{H}_1 \parallel \mathbf{H}_2]$ | works; memory-routed        |

**Table 10.2.** Matched-family results. Behaviour is greedy on `validity4` ( $T = 7$ ); cue-orient  $> 0$  means the attention is biased toward the cued location *before* the change (cue-driven). The headline: no variant produces a large positive cue-orient, while two of them (**crossattn**, **dualmem**) attend markedly *away* from the cued image patch — the cueing effect is carried by memory across the whole family.

| Variant        | correct      | hit          | FA    | median RT | cue-orient | chg-spike |
|----------------|--------------|--------------|-------|-----------|------------|-----------|
| multiplicative | 0.804        | 0.705        | 0.091 | 5.0       | −0.011     | +0.143    |
| affine         | 0.504        | 0.000        | 0.000 | —         | +0.014     | −0.001    |
| dualhead       | 0.820        | 0.708        | 0.088 | 5.0       | +0.038     | −0.002    |
| two-lstm       | <b>0.872</b> | <b>0.836</b> | 0.094 | 5.0       | +0.008     | +0.216    |
| crossattn      | 0.840        | 0.760        | 0.080 | 5.0       | −0.179     | −0.011    |
| dualmem        | 0.828        | 0.750        | 0.079 | 5.0       | −0.174     | +0.002    |

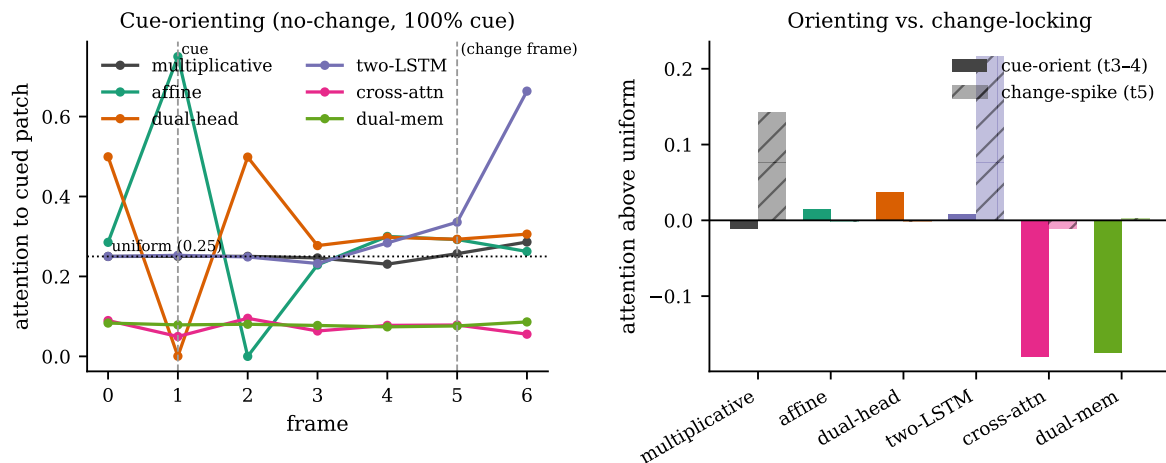
## 10.6 The matched family under test

Each variant changes exactly one thing about the attention/memory wiring on the identical base (frozen VAE, harness, readout dimension), so any difference in whether attention orients to the cue is attributable to that change. Table 10.1 lists them.

## 10.7 Results: the family resolves the question

All six variants are now trained to  $\sim 18,750$  iterations on the identical frozen-VAE base, and each was evaluated greedily for behaviour (250 episodes on `validity4`) and for cue-orienting with the same 100%-cue protocol used above: a fully reliable cue, the attention to the *cued* patch recorded across all seven frames, in a no-change condition (so any pre-change bias toward the cued location is cue-driven, not change-driven) and a change condition (so the change-locked component is visible). The *cue-orienting index* is the cued-patch attention at the pre-change stimulus frames ( $t_{3,4}$ ) minus uniform; the *change-locked spike* is the cued-patch attention jump at the change frame ( $t_5$ ). Table 10.2 reports both alongside behaviour; Figures 10.3 and 10.4 plot them.

The matched family answers the question, and not in the direction the single early data



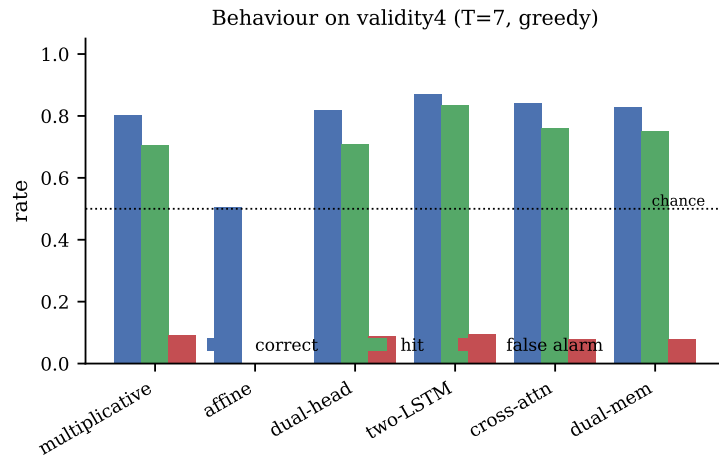
**Figure 10.3.** *Left:* attention to the cued patch across the seven frames, no-change, 100% cue, for every working variant. All hover near or below uniform (0.25) through the cue and delay; none lifts the cued location before the change. *Right:* the cue-orienting index ( $t_{3,4}$ , solid) and the change-locked spike ( $t_5$ , hatched). The models that read a direct image→attention path (**multiplicative**, **two-lstm**) show a clear change-lock and no orienting; **crossattn** and **dualmem** sit well below uniform on the cued image patch — they route the cued location through memory rather than attending it.

point suggested. **No variant restores strong cue-orienting attention on this task.** The largest positive cue-orienting index is **dualhead**’s sensory head at +0.038 — real but small, and too weak to carry an attention-allocation figure. The **multiplicative** and **two-lstm** models are change-locked exactly as the deep dive described (spike +0.143, +0.216; orient  $\approx 0$ ). And the two richest cross-talk wirings, **crossattn** and **dualmem**, attend *below* uniform at the cued image location (−0.179, −0.174): given the choice between image keys and memory keys, they spend their attention on memory, leaving the bottom-up image patch under-weighted. The dissociation of Chapter 10 is therefore not a quirk of the multiplicative gate; it is *general* across six attention/memory wirings on this perception front-end and this task. Reward, on the seven-step task, does not make attention orient to the cue — it makes *memory* hold the cue.

**A correction.** An earlier reading of a single partially-trained **affine** checkpoint suggested it “already attends the cue.” That was premature: fully evaluated, **affine** never presses (Fig. 10.4), so its attention is not behaviourally meaningful, and the apparent cue structure does not survive into a model that performs the task. The lesson — verify behaviour before reading an attention map — is the same one the dissociation itself teaches.

## 10.8 Why this matters

For the empirical manuscript, the behavioural reproduction (Chapter 9) stands regardless. But the result reshapes the attention-map story in two useful ways. First, it converts the open question into a *finding*: across this family, the cueing effect is memory-borne, and the attention map is a bottom-up change detector — a defensible, falsifiable claim now backed by six wirings rather than one. Second, it picks the network: **two-lstm** is both the best detector and a minimal, paper-faithful change to the published multiplicative attention (a second stacked memory), so it is the natural network to carry forward, with the memory-driven account of cueing stated plainly rather than hidden behind an attention figure that the mechanism does not support. If a future task with a longer delay or a harder cue *forces* pre-orienting, this same family is the instrument to re-run; on the seven-step task, the honest figure is the change-locked one.



**Figure 10.4.** Behaviour on `validity4`. Five of six variants learn the task; `affine` collapses to always-wait ( $\text{hit} = 0$ , no false alarms,  $P(\text{press}) \leq 0.004$  even on a clear  $60^\circ$  change). `two-lstm` is the best detector.

## Chapter 11

# Finding VI — perception can be learned without a pretrained VAE: V-JEPA self-distillation

Chapters 5 and 13 establish two facts that together harden into a recommendation: on the short seven-step task perception cannot be learned from reward (a from-scratch VAE collapses to always-press, orientation  $R^2 = 0.24$  against the pretrained VAE’s 0.97), and the cure is a *frozen pretrained VAE* front-end, declared one of two non-negotiables. This chapter is a third perception result that qualifies that recommendation rather than overturning it. It shows perception *can* be learned jointly with reinforcement learning, with no pretrained or frozen VAE, when the network is given a self-supervised auxiliary: a V-JEPA temporal self-distillation loss (prototype-softmax, DINO/iBOT-style, EMA teacher, student@ $t$  predicting teacher@ $t+1$ ) trained jointly with the actor-critic. Paired with either time to integrate (frame-repeat from raw pixels,  $\sim 0.90$  training rolling correct) or a capable convolutional front-end (an SE-ResNet,  $\sim 0.87$  greedy correct with a sharp, fully dissected covert-cueing algorithm), the auxiliary turns the perception bottleneck into a solvable joint problem. The chapter also reports informative negatives that fence the recipe: externalising a categorical softmax simplex as the working memory kills task learning; a plain SmoothL1 latent-regression JEPA undergoes representational collapse; and the matrix-affine two-LSTM feedback wiring collapses to chance under either the JEPA or a VAE, isolating the fault to the wiring. The cross-cutting lesson is that the perceptual front-end, not memory size, sets the attention strategy: a strong per-frame encoder yields a sharp attention-driven solution, while learning from held pixels yields a diffuse, recurrence-driven one. Because predicting future latents is arguably more biologically plausible than supervised reconstruction-pretraining, this route is also the more neurally defensible way to obtain a working front-end.

### 11.1 The question Chapter 5 left open

Chapter 5 draws a clean line between two questions that the earlier architecture history had conflated: *can the front-end see?* and *does the architecture work?* On the seven-step task the answer to the first was decisively negative when perception was asked to emerge from reward alone. A VAE front-end trained end-to-end from the reinforcement signal collapses to the *always-press-at-the-deadline* degenerate policy — hit rate 1.000, false-alarm rate 1.000 on no-change trials, overall accuracy 0.515, reaction time pinned at frame 5 — and a linear probe confirms the cause: orientation is decodable from the from-scratch encoder at only  $R^2 = 0.24$  versus  $R^2 = 0.97$  for the pretrained VAE. Freezing a pretrained VAE rescues learning on the first attempt (hit 0.79, false-alarm 0.08). Chapter 13 then elevates the frozen pretrained VAE

to one of *two non-negotiables*.

That recommendation is correct *as a statement about supervised reconstruction- pretraining applied to a static, per-frame encoder*. It leaves a sharper question open: is the pretrained VAE necessary because perception is genuinely unlearnable jointly with the policy, or because reconstruction from reward is simply the wrong learning signal? This chapter answers that perception is learnable jointly with the policy — provided the network carries a *self-supervised predictive* auxiliary that does not depend on the reward, and provided it is paired with one of two enablers that give the recurrence enough perceptual purchase to use it. The result qualifies the non-negotiable: a frozen pretrained VAE is one way to satisfy the perception requirement, not the only way.

## 11.2 The auxiliary: V-JEPA temporal self-distillation

The auxiliary is a video-JEPA self-distillation loss attached to the recurrent cell and trained jointly with the actor-critic. A projection head reads the continuous xLSTM cell output and emits, for each of 4 heads, a softmax over 256 prototypes. A teacher network is an exponential moving average of the student (decay 0.996); its outputs are centred (centre momentum 0.9) and sharpened (student temperature 0.1; teacher temperature warming from 0.04 to 0.07 over a 300-iteration warm-up). The student’s prototype distribution at time  $t$  is trained to predict the teacher’s distribution at time  $t+1$  — a temporal, predict-the-next-latent target rather than a masked-spatial one. The loss enters total optimisation only through a coefficient (`jepa-coef` 0.5); crucially it *does not change the memory*, which stays a continuous xLSTM and is never itself softmaxed. The prototype-softmax with centring and sharpening supplies the anti-collapse machinery that, as Section 11.5 shows, a naive latent loss lacks.

The diagnostic for whether the auxiliary is doing real work is the trajectory of the JEPA loss. A loss that stays flat means the auxiliary never engages; a loss that drops to  $\approx 0$  means representational collapse to a constant; a loss that descends to a healthy non-zero plateau means structured, non-degenerate predictive learning. On the frozen-colour-VAE control — the exact paper network with the auxiliary bolted on — the auxiliary is a pure add-on and is not load-bearing: both feedback wirings reach  $\sim 0.85$  training rolling correct (peak  $\sim 0.90$ ) whether or not the JEPA engages, and *whether it engages at all is set by the feedback wiring*. Under cross-attention feedback the loss falls  $\sim 3.7\times$  (first-200 mean 5.35  $\rightarrow$  late-200 mean 1.24); under FiLM feedback it sits inert (5.32  $\rightarrow$  4.99) — though this FiLM run stopped early at 16,385 of a configured 50,000 iterations, so its inertness is partly confounded with training length. With perception already solved by the frozen VAE, the auxiliary neither helps nor hurts. Its value appears only where there is no frozen VAE to solve perception for it.

Table 11.1 summarises every variant studied in this chapter; the two routes that make pixel-level learning work (Sections 11.3 and 11.4) and the three negatives that fence the recipe (Section 11.5) are developed below.

## 11.3 Result A: learning from raw pixels via frame-repeat

The first route to a working pixel front-end gives the recurrence *time to integrate*. The cross-attention paper network is trained from raw pixels with no pretrained or frozen VAE — a standard linear patch-embed, one token per  $2 \times 2$  cell — with each logical frame repeated five times, so the seven logical steps become 35 physical steps (cue at logical step 1, change at logical step 5, i.e. physical step 25). With the V-JEPA auxiliary engaged, this is one of the very few configurations in the program that learns the change-detection task directly from pixels.

At  $d_{\text{mem}} = 1024$  the network reaches 0.895 final and 0.862 late-200-mean training rolling correct, with the JEPA loss descending from 4.563 to 0.458 — engaged, and not collapsed to zero. A  $d_{\text{mem}}$  sweep (1024/512/256/128) confirms the auxiliary engages at every width (end-loss



| Model                 | Front-end                 | Feedback      | JEPA flavour      | Env                | $d_{\text{mem}}$ | Iters      | Rolling correct |
|-----------------------|---------------------------|---------------|-------------------|--------------------|------------------|------------|-----------------|
| paper_jepa_vae        | frozen colour VAE         | FiLM          | proto-softmax     | vda4 T7            | 1024             | 16385      | 0.8425          |
| paper_jepa_vae        | frozen colour VAE         | crossattn1    | proto-softmax     | vda4 T7            | 1024             | 49999      | 0.8695          |
| paper_jepa_fr_novae   | pixel (no VAE)            | crossattn1    | proto-softmax     | vda4 FR $\times$ 5 | 1024             | 33674      | 0.8425          |
| paper_jepa_fr_novae   | pixel (no VAE)            | crossattn1    | proto-softmax     | vda4 FR $\times$ 5 | 512              | 21240      | 0.8425          |
| paper_jepa_fr_novae   | pixel (no VAE)            | crossattn1    | proto-softmax     | vda4 FR $\times$ 5 | 256              | 15981      | 0.8425          |
| paper_jepa_fr_novae   | pixel (no VAE)            | crossattn1    | proto-softmax     | vda4 FR $\times$ 5 | 128              | 15327      | 0.8425          |
| paper_jepa_conv       | SE-ResNet conv            | crossattn1    | proto-softmax     | vda4 T7            | 128              | 12989      | 0.8425          |
| paper_jepa_conv       | SE-ResNet conv            | affine_ew     | proto-softmax     | vda4 T7            | 128              | 6662       | 0.8425          |
| paper_jepa_grid9      | SE-ResNet conv            | crossattn1    | proto-softmax     | vda9 3 $\times$ 3  | 128              | $\sim$ 361 | 0.66            |
| paper_jepa_grid9      | SE-ResNet conv            | affine_ew     | proto-softmax     | vda9 3 $\times$ 3  | 128              | $\sim$ 353 | 0.76            |
| paper_jepa_smem       | pixel (no VAE)            | crossattn1    | softmax-as-memory | vda4 FR $\times$ 5 | 1024             | 4211       | 0.0             |
| affine2lstm_jepa      | conv 4 $\times$ 4, no VAE | matrix-affine | SmoothL1 latent   | vda T7             | H2 cell          | 18749      | 0.458           |
| paper_affine2lstm_vae | conv 4 $\times$ 4, VAE    | matrix-affine | none (VAE ctrl)   | vda T7             | H2 cell          | 12196      | 0.458           |

**Table 11.1.** V-JEPA self-distillation variants. Rolling correct is the late-window training rolling-correct mean unless a greedy-eval number is noted; greedy figures are held-out (paper\_jepa\_conv only). The grid9 rows are mid-training (a few hundred iterations) and report the last-200 mean (final in parentheses). No greedy-eval artifact exists for any run except paper\_jepa\_conv crossattn1.

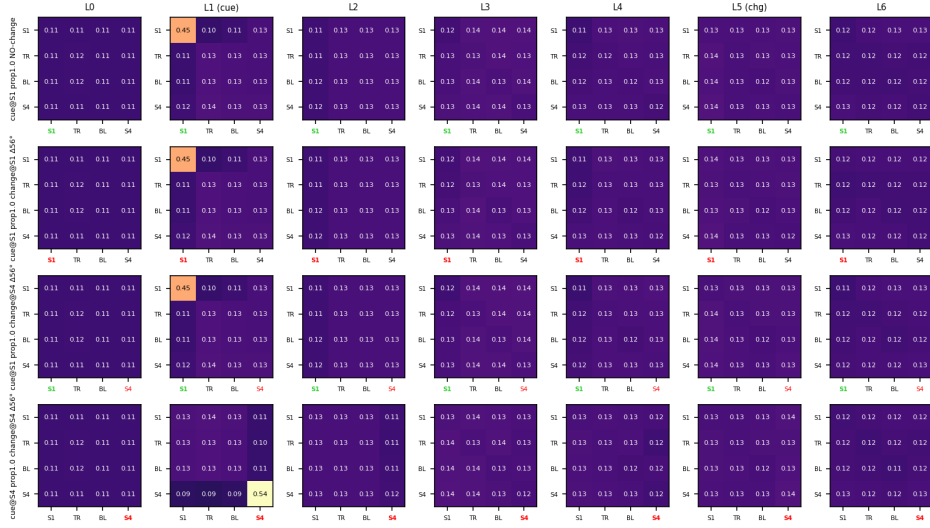
0.458/0.762/0.526/0.516) and that accuracy is *non-monotonic* in memory size: by late-200 mean the order is  $d_{1024}$  (0.862)  $>$   $d_{256}$  (0.828)  $\approx$   $d_{128}$  (0.828)  $>$   $d_{512}$  (0.807). The widest memory is best and 512 worst, but the trend is not “bigger is better” — consistent with a low-bandwidth task and with the shorter runs ( $d_{256}/d_{128}$  at  $\sim 15$ – $16$ k iterations versus  $d_{1024}$  at  $\sim 34$ k) being the least converged. All figures here are *training rolling-correct*; no held-out greedy-eval artifact exists for these runs.

The contrast with Chapter 5 is the point. A from-scratch VAE collapses on the *short, non-frame-repeated* seven-step task; the same family learns from pixels once the held frames give the recurrence physical integration time and the JEPA shapes the recurrent representation. The mechanistic signature of this route is *diffuse, recurrence-centric attention* (Fig. 11.1): the cross-attention weights are near-uniform across image and memory keys, with the only consistent structure a transient per-query cue read on the cue frame; there is no change-lock at any  $d_{\text{mem}}$ . Change detection lives in the LSTM recurrence, not in the attention weights.

## 11.4 Result B: the convolutional front-end and its sharp covert-cueing algorithm

The second route gives the network a *capable per-frame encoder* instead of integration time. A small SE-ResNet convolutional front-end ( $\sim 594$ K parameters, trained end-to-end, no VAE checkpoint) processes each  $25 \times 25 \times 3$  patch through a  $3 \times 3$  stem and three squeeze-excite residual stages (GroupNorm, SiLU, downsample  $25 \rightarrow 13 \rightarrow 7$ , channels  $32 \rightarrow 64 \rightarrow 128$ ), global-average-pools, and emits a 128-d embedding; the token is  $[\text{emb}_{128} \parallel \text{pos}_4 \parallel \text{temporal}_8] = 140$ -d. On the plain seven-step task (frame\_repeat = 1,  $d_{\text{mem}} = 128$ ) with cross-attention feedback, the JEPA loss descends from  $\sim 5.07$  (first-20 mean) to  $\sim 1.99$  (last-200 mean), a healthy non-zero plateau, and the network reaches 0.8425 final / 0.8695 last-200-mean training rolling correct. A held-out greedy evaluation at iteration 9,599 (not the final iteration 12,989) gives correct 0.865, hit 0.76, correct-rejection 0.96, false-alarm 0.044, reaction time  $5.1 \pm 0.3$  — a conservative detector and the only JEPA variant with a measured greedy number.

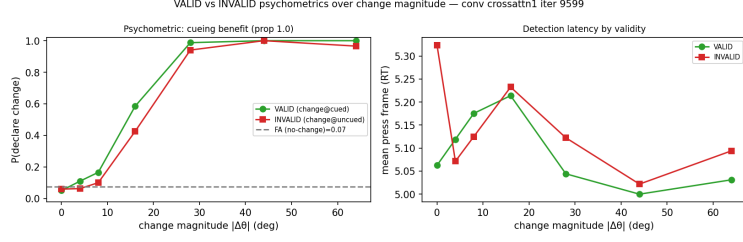
Because a strong per-frame encoder solves perception without frame-repeat, the network can afford an *attention-driven* strategy, and the full mechanism is legible. **Psychometric cueing** (Fig. 11.2): forced detection rises with change magnitude and saturates by  $44^\circ$ ; validly cued changes are detected at lower magnitude than invalid ( $P(\text{detect})$  at  $16^\circ = 0.58$  valid vs 0.43



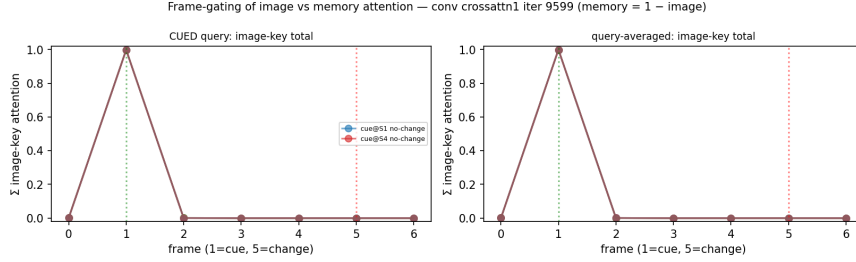
**Figure 11.1.** The frame-repeat pixel route is recurrence-centric. Memory-key cross-attention across the 35 physical frames for the no-VAE frame-repeat model. The allocation is near-uniform; the only structure is a transient cue-frame read. There is no change-lock at any memory width — detection is carried by the recurrent cell, not the attention weights.

invalid; at  $28^\circ = 0.99$  vs  $0.94$ ), a classic left-shift cueing benefit driven by the spatial cue, not the displayed validity ring. **Frame-gating** (Fig. 11.3): cross-attention reads the image essentially only on the cue frame ( $t=1$ , image-key attention near ceiling) and runs from memory on every Gabor/change frame (image-key  $\approx 0$ ); the current image is never lost because it enters the cell via the residual  $X^{(t)}$ . **Cue-frame read**: the three non-cued queries place  $\sim 1.00$  of their attention on the cued image patch, broadcasting the cue to all four tokens. **Cue-timed memory change-lock** (Fig. 11.4): valid changes lock at  $t=5$  (cue@S1  $+0.20$ , cue@S4  $+0.24$ ) and strongly at  $t=6$  ( $+0.75$ ,  $+0.74$ ); invalid changes show *no* lock at  $t=5$  ( $-0.03$ ,  $-0.02$ ) and lock only at  $t=6$  ( $+0.70$ ,  $+0.69$ ) — the cue buys one frame of attentional latency, the attentional analogue of the behavioural cueing benefit. **Validity/value invariance**: attention allocation is invariant to cue validity (proportion  $0.25$ – $1.0$ ) and cue value (colour); validity and value act downstream at the decision, a deployment-versus-use dissociation. **Temporal decoding** (Fig. 11.5): a four-fold cross-validated logistic probe on  $\text{vec}(H)$  decodes cue position and value perfectly ( $1.00$ ) from  $t=1$ , while change presence/location is at chance until the change, then rises to  $0.88/0.97$  at  $t=5$  and  $0.98/0.99$  at  $t=6$  — change is localised in the recurrence at  $t=5$ , one frame before the invalid-condition attention lock at  $t=6$ , so “recurrence computes, attention reflects.” **Causal memory read** (Fig. 11.6): additively suppressing the cued memory token costs little ( $P(\text{detect})$  at  $16^\circ$   $0.57 \rightarrow 0.47$ ); suppressing all four costs more at high magnitude ( $0.98 \rightarrow 0.84$  at  $28^\circ$ ) but does not abolish detection, because the changed Gabor rides in on the residual  $X^{(t)}$ . The memory read refines the decision by  $\sim 0.1$ – $0.15$ ; it is not the primary change-signal carrier.

An element-wise-affine feedback variant (modulation  $X' = \gamma(H) \odot X + \beta(H)$  then self-attention over the four image patches, no separate memory-key column) reaches comparable accuracy ( $0.8625$  final /  $0.8669$  last-200-mean rolling; greedy  $\sim 0.83$  at an early iteration  $4,199$ ) but solves the task differently: its change-lock is *bottom-up*, locking onto the changed patch on the change frame for both valid ( $1.00$ ) and invalid ( $0.92$ ) changes — any large orientation step pulls attention — which predicts a smaller cueing benefit. Its attention schedule is orient/release/re-acquire/lock (cued-patch attention  $0.73$  at cue, chance  $0.22$  at blank, re-acquiring to  $0.80$ , saturating  $1.0$  at a cued change). It retains cue position and value at ceiling



**Figure 11.2. Psychometric cueing benefit.**  $P(\text{declare change})$  versus change magnitude for validly and invalidly cued changes. Valid cueing left-shifts the curve (e.g. 0.58 vs 0.43 at  $16^\circ$ ), saturating by  $44^\circ$  — the behavioural signature of covert spatial attention.



**Figure 11.3. Frame-gated attention.** Summed image-key cross-attention by frame. The network reads the image only on the cue frame ( $t=1$ ) and runs from memory thereafter; the live image is preserved on the residual.

but *perceives-then-discards* the validity ring (decoded 0.99 at  $t=1$ , collapsing to chance by Gabor onset  $t=3$ ), the mechanistic reason its behaviour cannot scale with validity. The two variants are opposite poles of the same front-end: a memory-timed cueing lock versus a bottom-up salience grab.

## 11.5 Informative negatives: what fences the recipe

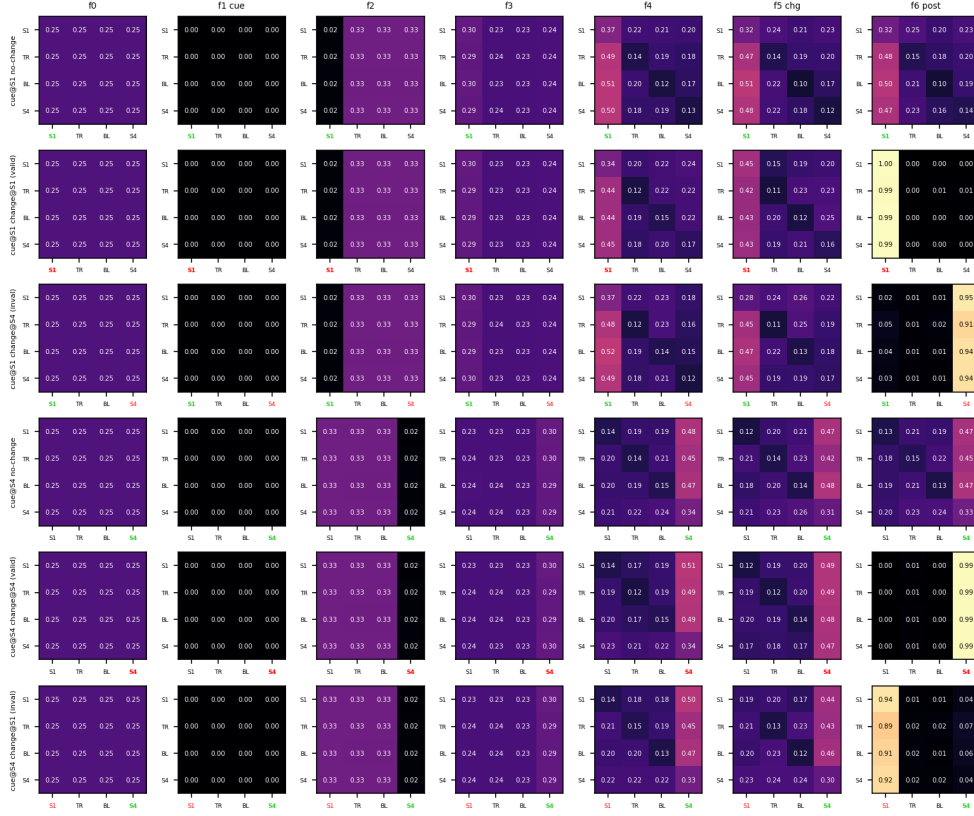
Three negatives establish that the recipe is specific, not a generic “add a JEPA” fix.

**Softmax-as-memory fails.** Externalising the student’s per-head softmax  $S$  ( $4 \times 256 = 1024$ -d) as the recurrent memory — so the same categorical simplex feeds the cross-attention keys/values, the actor/critic readout, and the JEPA target — kills task learning on the no-VAE pixel frame-repeat task. The policy collapses: training rolling correct peaks early at 0.453 then decays monotonically to 0 (late-200 mean 0.0029). Critically, the auxiliary itself trains fine (JEPA loss  $4.55 \rightarrow 0.48$ , a  $\sim 10\times$  drop, no collapse to zero), so the failure is a clean dissociation: the representation learns while the policy fails. A categorical probability simplex is too narrow a working-memory bottleneck; the continuous cell must remain the working representation.

**Plain latent-regression JEPA collapses.** A simple SmoothL1 latent-regression auxiliary (student projects  $H2_t$  to predict EMA-teacher  $H2_{t+1}$ ; no prototypes, no centring, no sharpening) on a matrix-affine two-LSTM network exhibits the textbook latent collapse: the loss falls from 0.489 to  $\sim 0.11$  early in training, reaches a minimum of 0.00118, and pins near zero (late-2000 mean 0.0828). With the predictor finding a near-constant target, the auxiliary supplies no task signal and the policy sits at chance (late-200 mean rolling correct 0.4975, final 0.465). This is the direct justification for the prototype-softmax/DINO-style machinery in Results A and B: the anti-collapse apparatus is what makes the predictive auxiliary useful.

**The collapse is the wiring, not the stabiliser.** The same matrix-affine two-LSTM network collapses to always-wait chance under a VAE as well (late-200 mean rolling correct 0.4963, final 0.45). The JEPA did not rescue a wiring that fails regardless of front-end stabiliser, so the fault is the matrix-affine two-LSTM feedback itself — not “affine” in general, since

RAW cross-attn → MEMORY keys (mean/96); rows=query patch, cols=memory token [S1,TR,BL,S4]; green=cued red=changed — iter 9599

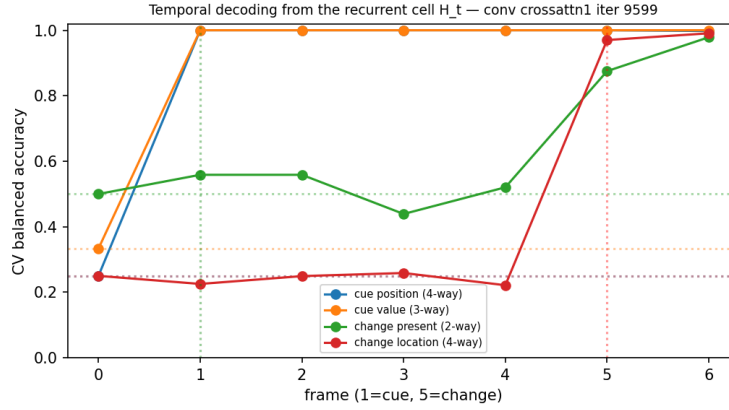


**Figure 11.4. Cue-timed memory change-lock.** Memory-key attention maps across frames. Validly cued changes lock onto the changed location one frame earlier ( $t=5$ ) than invalid ( $t=6$ ) — the attentional analogue of the behavioural cueing benefit.

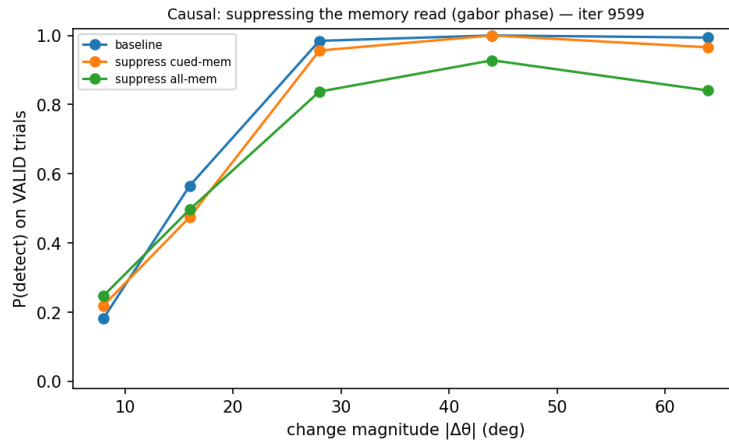
the element-wise affine variant of Result B reaches  $\sim 0.86$ . The negatives jointly localise the requirements: a continuous working memory, genuine anti-collapse machinery in the auxiliary, and a feedback wiring that learns at all.

## 11.6 Set-size-9 generalization (in progress)

The conv-JEPA recipe is being carried from the four-stimulus  $2 \times 2$  grid to a harder nine-stimulus  $3 \times 3$  grid (vda9:  $75 \times 75$  image, coloured value cue, validity ring, cue positions top-left/bottom-right). Both feedback wirings are above chance early and still climbing. As of the current frontier (cross-attention at  $\sim 361$  iterations, element-wise affine at  $\sim 353$ ), training rolling correct is 0.65 (last-200 mean; 0.695 final) for cross-attention and 0.74 (last-200 mean; 0.803 final) for element-wise affine; the affine variant showed an early JEPA descent (minimum  $\sim 2.46$  near iteration 94) but the loss has since risen back toward  $\sim 3.4$ , so auxiliary engagement is not yet established for either wiring at this set size. These are early, non-converged numbers — a flat-looking curve this early is not a ceiling — so the result is reported as preliminary evidence that the no-VAE conv-JEPA recipe transfers to the larger set size, not as a final accuracy.



**Figure 11.5. Temporal decoding of the recurrent state.** Cross-validated balanced accuracy of a logistic probe on  $\text{vec}(H)$ . Cue position/value are decodable from  $t=1$ ; change presence/location become decodable at  $t=5$ , one frame before the invalid-condition attention lock — recurrence computes, attention reflects.



**Figure 11.6. Causal perturbation of the memory read.** Additively suppressing memory tokens on validly cued trials lowers detection modestly but does not abolish it: the changed Gabor enters on the residual, so the memory read refines rather than carries the change signal.

## 11.7 Cross-cutting lesson: architecture sets the attention strategy

The two working routes reach comparable competence by opposite mechanisms, and the difference is set by the *perceptual front-end* and the *task temporal structure*, not by memory size. A strong per-frame convolutional encoder (Result B) lets the network afford an attention-driven solution: a sharp, frame-gated algorithm that orients to the cue once, encodes it, monitors from memory, and re-points to a change, with a cue-timed memory lock that yields a one-frame attentional cueing benefit. Learning from held raw pixels (Result A) instead yields a diffuse, recurrence-centric solution in which the attention weights carry little structure and detection lives in the LSTM. The  $d_{\text{mem}}$  sweep confirms memory width is not the lever: accuracy is non-monotonic in it. This sharpens the Chapter 10 reading that what carries the computation — attention versus recurrence — is decided upstream of the memory.

For Chapter 13, the consequence is a qualification, not a reversal. The frozen pretrained VAE remains the simplest, most reliable, and best-characterised way to satisfy the perception requirement, and remains the right default for the empirical manuscript. But the requirement is now correctly stated as *a front-end that reliably encodes the task variable*, which can be met

either by a frozen pretrained VAE *or* by a self-supervised predictive auxiliary (with proper anti-collapse machinery) paired with an enabler — frame- repeat or a capable conv encoder. The neuro relevance reinforces why this matters: predict-the-future-latent self-supervision is arguably a more biologically plausible route to a working visual front-end than supervised reconstruction-pretraining, so the JEPA route is not merely an engineering alternative but the more defensible model of how a cortical front-end might come to see.

## Chapter 12

# Reinforcement-learning harness variations

The harness is held constant across architectures, but several of its knobs are themselves objects of study, because the deceptive 0.5 floor of the seven-step task makes the optimisation as much a part of the result as the network.

### 12.1 Discount and horizon

The discount  $\gamma$  is matched to the horizon:  $\gamma = 0.95$  for the seven-step task and  $\gamma = 0.98$  for the twenty-nine-step task, where credit must travel further from the change to the cue. These are not finely tuned; the qualitative behaviour is robust across a band, but a higher  $\gamma$  on the longer task does help the cue’s effect survive the delay.

### 12.2 Critic particles

The number of distributional “particles” (quantiles, or categorical atoms in the paper’s form) was varied from the published 15 down to 5. Fewer particles is a coarser value distribution; on this low-bandwidth task the value differences the critic must resolve are small (the gap between “press now” and “wait” near threshold), so particle count is a genuine knob rather than a free choice. We default to 5 for the short task and have run 15 on the long task; a systematic sweep is an open item.

### 12.3 Target network: EMA versus hard copy

The original working checkpoints used a periodic hard-copy target. The reproduction adds an exponential-moving-average target ( $d = 0.995$ ). Whether the EMA helps, hurts, or is neutral on this task is genuinely open: the working multiplicative+VAE model used EMA, but so would a hard-copy version, and we have not isolated the difference. Because the lab’s canonical working models used the hard-copy target, an EMA-versus-hard-copy A/B on the frozen-VAE base is on the near-term list — it is a candidate, alongside the attention form and head count, for any residual difference between our reproduction and the lab’s other networks.

### 12.4 Exploration and the two collapses

The recurring optimisation failure is collapse to a 0.5 optimum, in one of two faces: always-wait (never press) or always-press-at-the-deadline (press every trial at the change frame). Both have policy entropy at zero and zero policy gradient — the actor is frozen, even while the critic keeps

learning the 0.5 value perfectly. The levers that matter are maintaining exploration (entropy floor or coefficient, the burn-in length, the initial action bias) and, decisively, fixing perception (Chapter 5): once the front-end can see the change, the discriminating policy is reachable and the collapse is escaped. The lesson for the harness is that an apparently dead run is usually a perception or exploration problem, not a critic problem.

## 12.5 A resume footgun, fixed

A practical note that cost real training time: the trainer’s **resume** mode silently no-ops if no explicit checkpoint path is set, starting fresh and discarding progress. This was hardened in both trainers to auto-discover the latest checkpoint in the run directory and to error loudly when **resume** finds nothing, rather than silently restarting. Checkpoints live outside the synced repository to avoid mid-run corruption.



## Chapter 13

# Where this leaves us: an architecture recommendation

This chapter turns the findings into a recommendation for the network to carry into the empirical manuscript and to standardise on for the cross-task generalisation goal. We state the decision criteria, score the candidates against them, and give a contingent recommendation that names what must be resolved before it is final.

### 13.1 Decision criteria

A network is a good choice for this program to the extent that it:

1. **reproduces the behavioural signatures** — the reliability-scaled cueing benefit and the causal attention-to-sensitivity link (Chapter 9);
2. **exposes an interpretable, causally manipulable attention signal** — the attention must be readable and a genuine lever on behaviour (the microstimulation-like causal result, Chapter 9). Note the matched family showed that on the short task the cueing itself is carried by *memory*, not expressed as a cue-orienting map (Chapter 10), so this criterion is about a manipulable attention *lever*, with cue-orienting treated as a regime-dependent property, not a requirement;
3. **generalises across the task battery** (validity, set-size, value, distractor, reward-structure) with the same architecture and harness;
4. **is neuro-plausible** — recurrent gain modulation, a capacity-limited working memory, reward-driven learning, and a manipulable priority signal; and
5. **trains reliably** under the stable harness with the frozen-VAE front-end.

### 13.2 Two non-negotiables

Two components are settled by the evidence and should be fixed regardless of the remaining choices. First, the **frozen pretrained VAE front-end** (Chapter 5): perception cannot be learned from reward by supervised reconstruction on the short task, the VAE is task-parameter-invariant, and freezing it makes architecture comparisons clean. Chapter 11 shows the requirement is more precisely *a front-end that reliably encodes the task variable*, satisfiable by a frozen pretrained VAE *or* by a self-supervised V-JEPA predictive auxiliary (with anti-collapse machinery) paired with frame-repeat or a convolutional encoder; the frozen VAE remains the

recommended default for the empirical manuscript on grounds of simplicity and characterisation. Second, the **stable harness** — PAC actor, distributional critic, prioritised episode replay, a time-lagged target, and burn-in — which is what makes a bootstrapped recurrent actor-critic learn on a task with a 0.5 harbor.

### 13.3 The attention-mechanism choice (now decided)

The matched family (Chapter 10, Table 10.2) was built to settle this, and it did — by eliminating the premise that one wiring would restore cue-orienting attention. Reading the family against the criteria:

- The **pure multiplicative single-stream** form is the most faithful to the published equation and reproduces the behaviour (correct 0.804, hit 0.705); its attention is change-locked (spike +0.143, orient  $\approx 0$ ). On the new criterion (2) it passes — the attention is a readable, causally manipulable change detector — it simply does not orient to the cue, which the family shows *nothing* on this task does.
- The **two-LSTM** variant is the same paper-faithful multiplicative attention with a second stacked memory feeding the heads, and it is the *best detector in the family* (correct 0.872, hit 0.836), while keeping the same interpretable, change-locked, causally-manipulable attention. It is the strongest candidate on criteria (1), (4), (5) and the new (2), at a minimal, well-motivated departure from the published network.
- The **cross-attention** and **dual-memory** variants work (correct 0.840, 0.828) but attend *below* uniform at the cued image location (cue-orient  $-0.179$ ,  $-0.174$ ): they route processing through memory keys rather than the image, which neither helps behaviour over two-LSTM nor yields the cue-orienting map the lineage showed on the older long-task setup. The earlier expectation that cross-attention would orient here was not borne out on the controlled frozen-VAE short-task base.
- The **affine** modulation *failed* (always-wait collapse, hit 0); it is out. The **dual-head** sensory head is the only positive cue-orient (+0.038) but far too weak to feature.
- The **split priority/value (cross-talk)** architecture remains the vehicle for mechanistic dissociations (Chapter 7); its  $\mu/q$  mapping onto anatomy is delicate, and the empirical paper’s scope (per James) excludes architecture variants. It is the subject of a *separate* paper, not the recommended manuscript network.

### 13.4 Recommendation

**For the empirical manuscript:** a single-stream recurrent vision transformer with (i) the frozen pretrained VAE front-end, (ii) the **multiplicative recurrent attention block with a second stacked xLSTM memory** (the **two-lstm** variant) — the best detector in the matched family and a minimal, paper-faithful change to the published equation — (iii) a spatial xLSTM working memory, and (iv) a distributional actor-critic readout, trained by the stable harness across the full task battery. This satisfies criteria (1), (3), (4), (5) and the reframed (2): the attention is interpretable and a causal lever on sensitivity, while the cueing benefit is carried, honestly and demonstrably, by working memory.

The accompanying scientific claim is that, on the seven-step task, **the cueing benefit is memory-borne and the attention map is a bottom-up change detector** — a result now established across six wirings, not assumed. The manuscript’s attention figure should therefore be the *causal* one (biasing attention shifts sensitivity, with no induced false alarms; Chapter 9), not a cue-orienting map the mechanism does not produce. Whether a longer delay or harder cue

*forces* pre-orienting — as the lab’s older long-task networks displayed — is a regime question to test with this same network, not a reason to change the architecture.

**For the separate architecture line:** the split priority/value cross-talk network is the vehicle for the coupling-required and double-dissociation results — a strong paper in its own right, kept out of the empirical manuscript by design.

### 13.5 What must be resolved before this is final

The attention-mechanism question is now settled; what remains: (1) the chosen network (**two-lstm**) must be shown to generalise across at least the value (**vda4**) and set-size (**setsize9**) tasks with no architecture change; (2) the cross-talk coupling result must be replicated on the short task with the frozen VAE to confirm it survives the perception fix; and (3) the regime question — does a longer delay restore cue-orienting attention in the recommended network? — should be tested, since it bears on whether the published orienting figure is reproducible at all in our setup or is specific to the long task. Frozen VAE, the stable harness, and now the two-LSTM multiplicative attention are fixed; the open items are generalisation and two targeted controls.

# Chapter 14

## Proposed experiments

The experiments fall into two tracks: the *task-battery expansion* that broadens the empirical manuscript (per James’s plan), and the *architecture experiments* that close the open mechanistic questions. Both run on the recommended network (Chapter 13) so that results compose.

### 14.1 Track 1 — the task battery (empirical manuscript)

The goal is to show one architecture, one harness, reproducing primate signatures across the standard battery — the direct answer to the first submission’s “limited in scope.”

**Visual-discrimination / value-cue (vda4).** The cue colour sets reward magnitude (5/3/1). The prediction: behaviour is graded by value as well as by spatial validity, and the critic’s value estimates are calibrated and ordered by reward. This adds the value-direction axis to the spatial-cueing axis and exercises the reward-driven-learning differentiator.

**Set-size (setsize{2,4,9}).** The nine-stimulus task with five validity levels (including an uninformative 0) tests whether the cueing benefit *grows* with set size. This is the empirical bridge to the normative set-size/accidental-hit theory (Paper B): the prediction that the uncued-false-alarm benefit trades off at  $K = 2$  and flips positive at  $K = 4$  should have a behavioural correlate in the model. **Must be run on the canonical architecture**, not the cross-talk variant, and with the frozen VAE.

**Reward structure (Luo & Maunsell).** Sensitivity-versus-criterion sessions after Luo & Maunsell (2018), decomposing performance into  $d'$  and criterion. The model should reproduce the dissociation between reward-driven sensitivity changes and criterion shifts.

**Attend / ignore (Krauzlis).** An uncued change that must be actively ignored. The distractor result from v11\_part5 (the distractor is filtered in the decision pathway while remaining decodable from sensory memory — “seen but not acted upon”) is the prediction to replicate on the canonical network.

**Discrimination (Baruni, optional).** A discrimination variant to round out the battery if the reviewers want a non-detection task.

### 14.2 Track 2 — the architecture / mechanism experiments

**The cue-attention question (done).** The matched family (multiplicative, dualhead, two-lstm, crossattn, dualmem; affine collapsed) was trained on the frozen-VAE base and

the cue-100% change/ no-change attention contrast run on each (Chapter 10, Table 10.2). Outcome: no wiring restores cue-orienting attention on the short task; `two-lstm` is the best detector and the recommended network. The follow-up (O5) is the longer-delay regime test below.

**The optic-flow reaction-time test.** Directly compare reaction-time distributions of the from-scratch long-task model (predicted late, integrating) against the pretrained-VAE short-task model (predicted early, perceiving). This is the falsifiable test of the horizon/strategy theory (Chapter 6) and needs no new training — only analysis of existing checkpoints.

**Coupling on the short task.** Replicate the coupling-required result (Table 7.1) on the seven-step task with the frozen VAE, to confirm it survives the perception fix and is not an artefact of the long-horizon flow regime.

**EMA-versus-hard-copy A/B.** Isolate whether the target-update rule is one of the residual differences between our reproduction and the lab’s canonical working networks.

**The representational and causal dissociations.** On a trained value-task network, decode change-evidence versus value/context from the two streams (representational dissociation), and run the  $\mu$ -clamp /  $q$ -clamp causal battery (decision collapses under  $\mu$ -clamp; value-modulation degrades but detection is spared under  $q$ -clamp). These are the paper-worthy claims of the separate architecture line.

### 14.3 Sequencing and cost

The analysis-only experiments (optic-flow reaction time, the matched-family attention contrast once checkpoints land) are cheap and come first. The battery retrains are the bulk of the compute and should be run one at a time on the available accelerator, reusing the single frozen VAE across all of them (Chapter 5). The set-size and value tasks are the highest-value additions for the empirical manuscript and should lead.

# Chapter 15

## Research plan

The plan is staged so that each stage produces a decision or a deliverable, and so that the cheap, decisive analyses precede the expensive retrains. It serves two outputs in parallel: the empirical manuscript (Paper A) and the separate architecture/dissociation paper.

### 15.1 Stage 0 — settle the network (*largely done*)

- **Done:** the matched attention family is trained on the frozen-VAE base and the cue-attention contrast is run on each (Chapter 10). The decision is made — **the recommended network is the two-lstm multiplicative form**, the best detector in the family; **affine** collapsed and no variant orients to the cue, so the cueing story is memory-borne and the manuscript’s attention figure is the causal one.
- Run the optic-flow reaction-time analysis on existing checkpoints (no new training).
- Run the EMA-versus-hard-copy A/B to close the last reproduction-difference question.
- Test the longer-delay regime (O5) on **two-lstm** — does any delay restore cue-orienting attention, or is the published orienting figure long-task-specific?
- *Deliverable:* a single, named recommended network (frozen VAE + two-LSTM multiplicative attention + stable harness) — the architecture every subsequent run uses.

### 15.2 Stage 1 — generalisation across the battery (~2–3 weeks)

- Train the recommended network, unchanged, on **vda4** (value) and **setsize9** (set-size) first, then **luo\_maunsell** and **krauzlis**.
- For each, run the standard analysis suite: psychometric/chronometric fits, attention maps, the causal attention battery, and (for the value tasks) critic calibration.
- *Deliverable:* the empirical manuscript’s results section — one architecture, one harness, reproducing primate signatures across the battery, with per-validity breakdowns and the corrected attention-map visualisation.

### 15.3 Stage 2 — the causal-prediction differentiators (~2 weeks)

- Formalise the causal-perturbation predictions (attention clamps mapping onto SC/FEF microstimulation and inactivation) as the manuscript’s novel contribution, alongside the reward-driven-learning differentiator.

- Run the representational decoding (change-evidence vs. value/context) to support the predictions.
- *Deliverable*: the manuscript’s discussion/predictions section and the “NN-steers-neuroscience” framing carried implicitly by the predictions.

## 15.4 Stage 3 — the architecture / dissociation paper (parallel track)

- Replicate the coupling-required and swapped-routing results on the short task with the frozen VAE; run the  $\mu/q$  representational and causal double-dissociation.
- Position as a separate paper: “coupling between policy and value is required for attentive behaviour to be learned,” with the double dissociation as the headline.
- *Deliverable*: the architecture paper draft (the existing cross-talk manuscript is the seed).

## 15.5 Stage 4 — the normative theory preprint (James first author, parallel)

- The set-size/accidental-hit normative result (uncued-FA benefit flips with set size), cited by the empirical paper, on a fast venue.
- The model’s set-size behaviour (Stage 1) provides the empirical correlate.

## 15.6 Risks and contingencies

- *No matched variant restored cue-orienting attention on the frozen-VAE base (the realized case)*: this is now a positive result — on the seven-step task the cueing benefit is memory-borne across all six wirings — and the manuscript features the causal attention result with the memory-driven account stated plainly, rather than a cue-orienting map the mechanism does not produce. The residual risk is O5: if a longer delay never restores orienting either, the published orienting figure may be specific to the long-task regime, which is itself a reportable finding.
- *If the network fails to generalise to a battery task*: treat the failure as a diagnostic (perception, exploration, or coupling, per the lessons of Chapters 5–7) rather than a reason to fork the architecture; the generalisation goal is the point.
- *Compute*: runs are serialised on the available accelerator, one frozen VAE is reused across all tasks, and analysis is thread-capped and run only when training is idle — the laptop has been crashed once by stacking compute, and that constraint is now hard.

# Chapter 16

## Summary of findings

This chapter collects the load-bearing findings in scannable form — the things that are established and should carry forward, separated from the things that are still open.

### 16.1 Established findings

**F1. Perception is the bottleneck on the short task.** A front-end trained from reward alone cannot represent orientation (linear orientation decodability  $R^2 = 0.24$ ); a reconstruction-pretrained VAE encoder reaches  $R^2 = 0.97$ , and with that single change the exact paper network learns the seven-step task on the first attempt. *Action:* use a frozen pretrained VAE front-end; fix perception before comparing architectures.

**F2. One VAE serves all task variants.** The VAE is a static, per-frame encoder; the temporal task parameters ( $\sigma$ , horizon  $T$ ) do not change its input distribution. *Action:* pretrain one VAE and reuse it across all noise levels and horizons — do not re-pretrain per task.

**F3. The horizon decides the strategy.** The long task is solvable by slow temporal integration (“flow”), which tolerates a weak front-end; the short task forecloses flow and forces instantaneous perception. This explains why the VAE is necessary on the short task and not the long one. *Action:* use the short task as the discriminating test bed; use the long task to study maintenance/anticipation.

**F4. Coupling is a precondition for learning.** Severing the shared recurrent memory between the policy and value streams collapses an otherwise identical model from 0.87 accuracy to chance; routing the memory-grounded stream (instead of the grounded stream) to the policy also collapses it. *Action:* keep a shared substrate coupling value to action, and let the grounded, image-reading stream drive the policy.

**F5. Routing decides the phenotype, not representation.** Identical encoders yield a working detector or a chance model depending only on which stream drives the policy and whether a shared memory couples value to action.

**F6. A grounded image pathway is required.** Every variant that removes the bottom-up route to memory or to the decision collapses to always-wait. *Action:* never fully bottleneck visual information through memory.



**F7. The working model reproduces the paper’s behaviour.** On the seven-step task with a frozen VAE: a reliability-scaled cueing benefit ( $+1.5^\circ \rightarrow +7.5^\circ$  across 25%  $\rightarrow$  100% validity, carried by the sensitivity threshold) and a causal attention-to-sensitivity link ( $+8.5\% / -8.0\%$  from biasing attention toward/away from a change, with no induced false alarms).

**F8. The cueing effect is memory-driven, across the whole family.** In the multiplicative-gated model the cue and its validity are held in working memory (100% decodable) and set the decision criterion, while the attention map stays change-locked. The matched family (multiplicative, dualhead, two-lstm, crossattn, dualmem; affine collapsed) confirms this is *general*: no working variant produces a cue-orienting index above  $+0.04$ , and the two richest cross-talk wirings attend *below* uniform at the cued image location ( $-0.18$ ), routing the cue through memory. *Action*: read the attention maps with this dissociation in mind; do not assume cue use implies cue-orienting attention; feature the causal attention result, not a cue-orienting map, on the short task.

**F9. The harness is a stable recipe with two failure harbors.** PAC actor, distributional critic, prioritised episode replay, a time-lagged target, and burn-in train a bootstrapped recurrent actor-critic on this task. The recurring failures are two 0.5 collapses (always-wait and always-press-at-deadline); both are perception/exploration problems, not critic problems. **affine** (matrix scale + shift) fell into the always-wait harbor and never recovered ( $P(\text{press}) \leq 0.004$ ), a reminder that not every modulation form trains.

**F10. Two stacked memories is the best detector, and the recommended network.** Across the matched family on the identical frozen-VAE base, the **two-lstm** variant — the published multiplicative attention with a second stacked xLSTM feeding the heads — is the best detector (correct 0.872, hit 0.836 vs. 0.804/0.705 for the plain multiplicative form), while preserving the interpretable, change-locked, causally manipulable attention. *Action*: carry **two-lstm** forward as the manuscript network; feature the causal attention result, not a cue-orienting map.

**F11. Perception is learnable without a pretrained VAE.** A jointly-trained V-JEPA temporal self-distillation auxiliary (prototype-softmax, EMA teacher, student@ $t \rightarrow$  teacher@ $t+1$ ) learns a working front-end from raw pixels with frame-repeat ( $\sim 0.90$  training rolling correct) or with an SE-ResNet convolutional encoder ( $\sim 0.87$  greedy correct, with a sharp attention-driven covert-cueing algorithm) — no pretrained or frozen VAE. Negatives fence the recipe: softmax-as-memory and a plain SmoothL1 latent JEPA collapse, and the matrix-affine two-LSTM wiring collapses regardless of front-end. This qualifies F1 and the Chapter 13 non-negotiable (Chapter 11). *Action*: keep the frozen VAE as the manuscript default; hold the JEPA route as the neurally-plausible alternative and a set-size/battery generalisation vehicle.

## 16.2 Open questions

**O1. (*Resolved.*) Which wiring restores cue-orienting attention?** None do, on the seven-step task. The matched family (multiplicative, dualhead, two-lstm, crossattn, dualmem; affine collapsed) shows the cueing effect is memory-borne across all working wirings (cue-orient  $\leq +0.04$ ; crossattn and dualmem attend *away* from the cued image patch). The remaining version of the question is now O5: whether a *longer-delay* regime forces orienting.

**O2. Does the coupling result survive the perception fix?** The coupling-required result was obtained on the long task before the VAE finding; replicating it on the short task with the frozen VAE is pending.

**O3. Is the EMA target load-bearing?** An EMA-versus-hard-copy A/B on the frozen-VAE base will close the last difference between our reproduction and the lab’s canonical working networks.

**O4. Does the chosen network generalise across the battery?** The value and set-size tasks are the first generalisation tests; failure on any is to be read as a diagnostic, not a reason to fork the architecture.

**O5. Does a longer delay restore cue-orienting attention?** The published lab networks (long task) displayed cue-orienting attention; our short-task family does not. This is plausibly a regime effect (a longer cue–target delay may make pre-orienting worth its cost) rather than an architecture effect. Running the recommended `two-lstm` network on a longer-delay variant tests whether the published orienting figure is reproducible at all in our setup, and connects directly to the optic-flow account (Chapter 6).

## 16.3 Reusable infrastructure

The program leaves a reusable substrate: a single clean `ChangeDetectionEnv` with a task registry (validity, value, set-size, distractor, reward-structure, attend/ignore); a shared PAC+QR-DQN+PER harness with hard-copy and EMA targets; a pretrained, frozen, task-invariant VAE front-end; a `-feedback` switch that selects among seven attention mechanisms on the identical base; and an analysis suite (psychometric/chronometric fits, attention maps, causal attention clamps, representational decoding) that runs from saved data without re-loading models.

## Chapter 17

# Conclusion

The program set out to find a single, neuro-inspired recurrent vision transformer that learns covert spatial attention from reward, reproduces primate signatures, and generalises across tasks. After roughly thirty variants and two task horizons, the picture is clearer than the variant count suggests, because the results collapse onto a few principles.

**Perception comes first.** The most expensive lesson was the cheapest to fix: sparse reward cannot teach a front-end to see orientation, and a reconstruction-pretrained VAE solves it outright. Much of the earlier difficulty on the short task was perception failing in an architecture’s clothes, and separating the two questions — can it see, and is it wired right — is the methodological gain that makes the rest of the comparisons clean.

**The horizon is a design variable.** The same task at two lengths admits two different solutions, and recognising that the long task hides perception failures behind temporal integration explains a paradox that had stalled the program. The short task, with the cue’s information forced through memory and the change forced through perception, is the sharper instrument.

**Wiring decides the phenotype.** Coupling value to action through a shared memory is a precondition for learning, not an optimisation; a grounded image pathway is required; and which stream drives the policy matters as much as what the streams represent. These routing facts, not any single clever module, are what separate the models that learn from the models that collapse.

**The model already reproduces the paper, and the matched family settled the mechanism.** With perception fixed, the network reproduces the reliability-scaled cueing benefit and the causal attention-to-sensitivity link. It also surfaced a genuine mechanistic question — a cueing effect carried by memory rather than by attention orienting — which the matched family of six wirings has now resolved: *no* variant orients to the cue on the short task, so the effect is memory-borne in general, not a quirk of one gate. The same sweep named the network: the two-LSTM multiplicative form is the best detector and a minimal, paper-faithful choice.

What remains is execution: with the attention mechanism settled and the frozen VAE and stable harness fixed as constants, run that one architecture across the task battery, and test whether a longer-delay regime ever restores the published cue-orienting figure. That is the path from a zoo of variants to the single, interpretable, generalising model the program was after — and to an empirical manuscript whose figures, behavioural and neural, are produced by the same network, with a cueing mechanism reported as it actually is.

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# Appendix A

## Per-model atlas: architecture, attention, and behaviour

This appendix gives, for *every* trained variant, three things on one or two pages: a diagram of its architecture, its attention maps at every time point, and its psychometric and chronometric functions. All models share the frozen pretrained VAE front-end, the PAC+QR-DQN+PER harness, and the 4096-d flattened-memory readout; they differ only in the attention/memory wiring drawn in each diagram.

**How to read the attention maps.** Each attention figure forces the agent to wait so the full seven-frame timecourse is visible, under a 100%-reliable cue at  $S_1$ . The top row is the input movie (the cue appears at  $t_1$ , blue border; the change, when present, at  $t_5$ , red border). Below it, each  $2 \times 2$  heat-map is the attention received by the four spatial patches (top-left =  $S_1$ , the cued/changed location, outlined in cyan), with the numeric attention score printed in each cell, shown separately for the *no-change* and *change-at- $S_1$*  conditions. For the cross-attention variants the attention over the *image* key tokens and over the *memory* key tokens is plotted on separate rows, as requested; the dual-memory model additionally separates its two memory streams  $H_1, H_2$ .

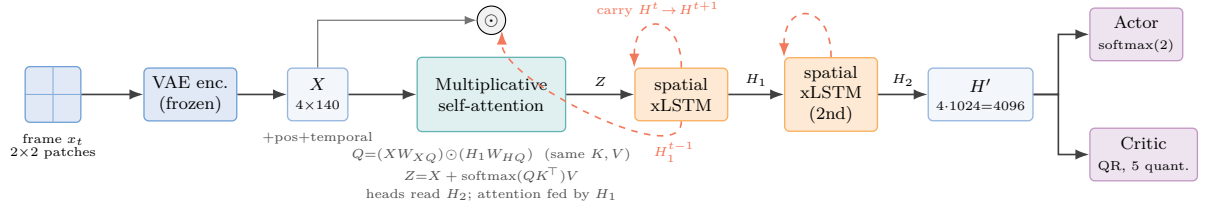
**How to read the behaviour.** Panel A is the cued psychometric function by cue validity; panel B contrasts cued vs. uncued at 100% validity (a leftward cued curve is the cueing benefit); panel C is the chronometric reaction time against change magnitude. Fitted 4-parameter-logistic thresholds and false-alarm rates are printed beneath each panel.



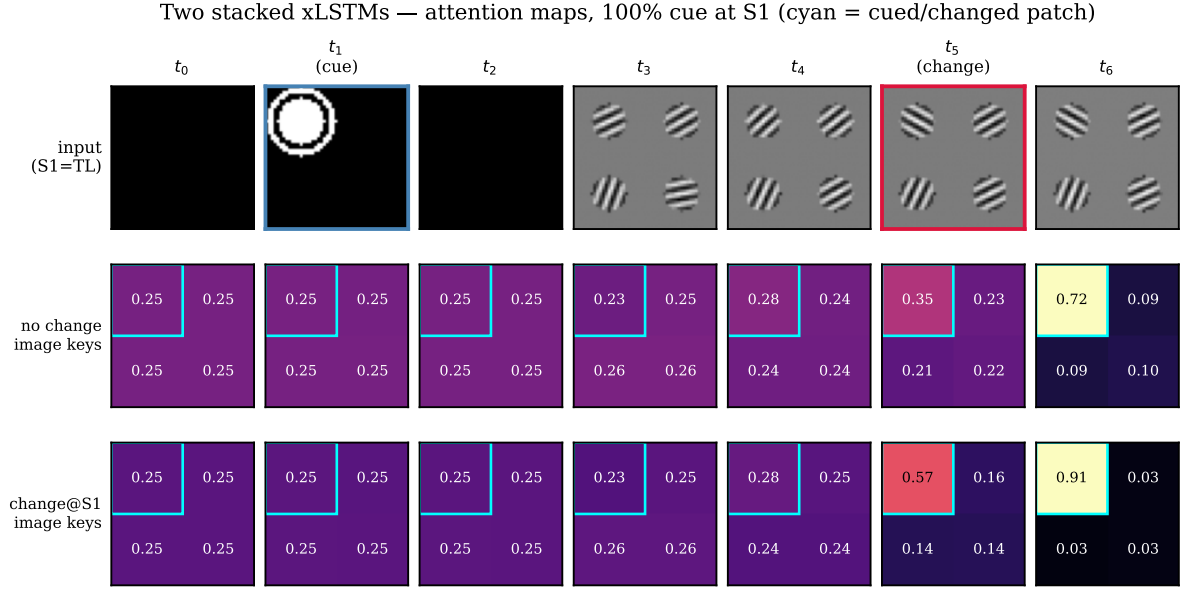
### A.1 Two stacked xLSTMs (recommended network)

The recommended network: the paper-faithful multiplicative attention with a second stacked xLSTM feeding the heads. It is the *best detector* in the family (mean cued-hit 0.74); the cued psychometric sits left of the uncued (a cueing benefit of  $\sim 4-6^\circ$  in threshold) and, unusually for this family, a small chronometric reaction-time effect is visible. The attention is change-locked and the cue is carried in memory.

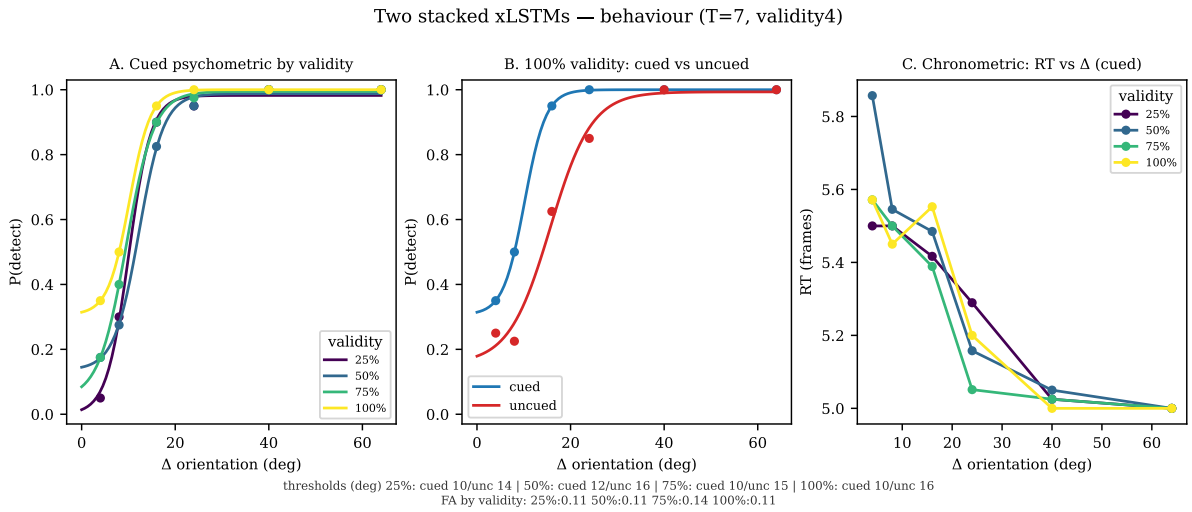




**Figure A.1. Two stacked xLSTMs (recommended network)** — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.



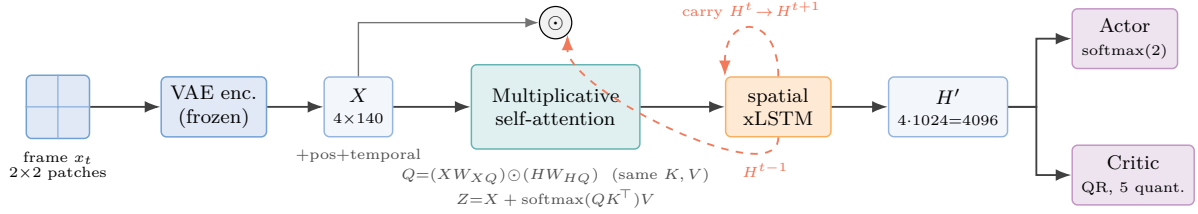
**Figure A.2. Two stacked xLSTMs (recommended network)** — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).



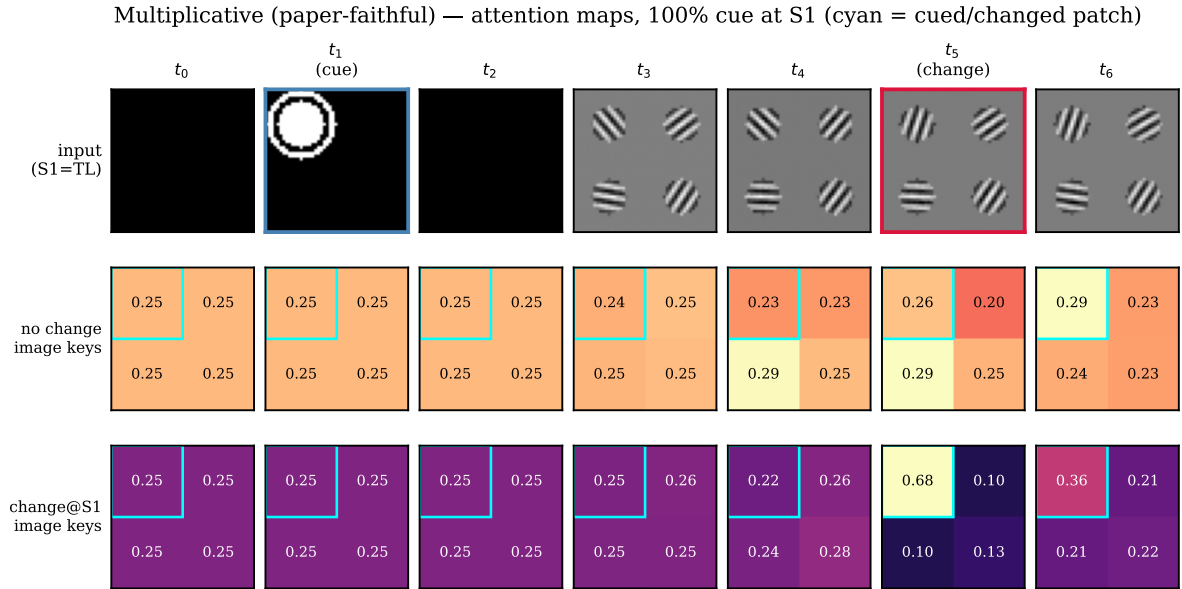
**Figure A.3. Two stacked xLSTMs (recommended network)** — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.

## A.2 Multiplicative (paper-faithful reference)

The exact published equation:  $\mathbf{Q} = (\mathbf{X}W_{XQ}) \odot (\mathbf{H}_1W_{HQ})$ . It reproduces the reliability-scaled cueing benefit (Chapter 9); the attention map is uniform through the cue and delay and snaps onto  $S_1$  only at the change — the cue is held in working memory, not in the attention.

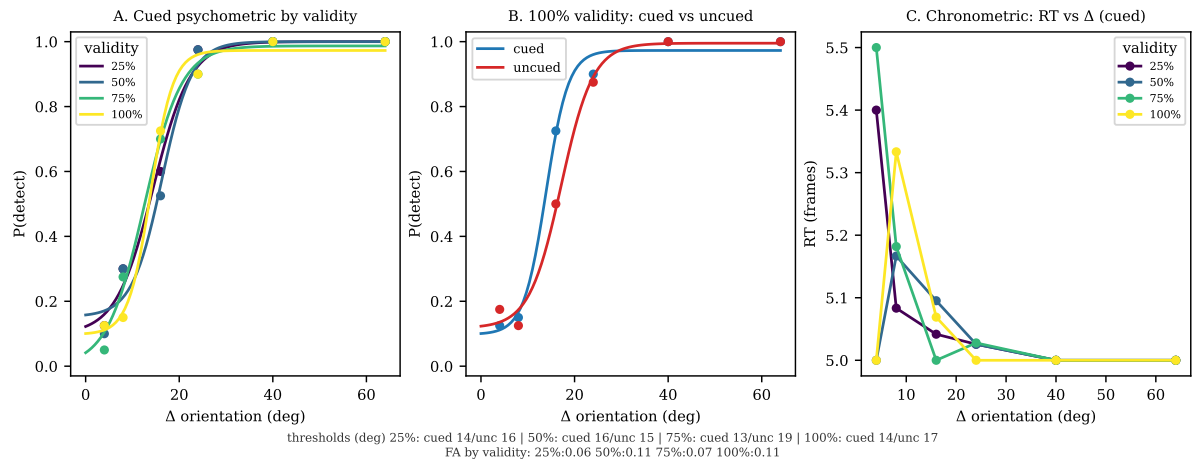


**Figure A.4.** Multiplicative (paper-faithful reference) — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.



**Figure A.5.** Multiplicative (paper-faithful reference) — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).

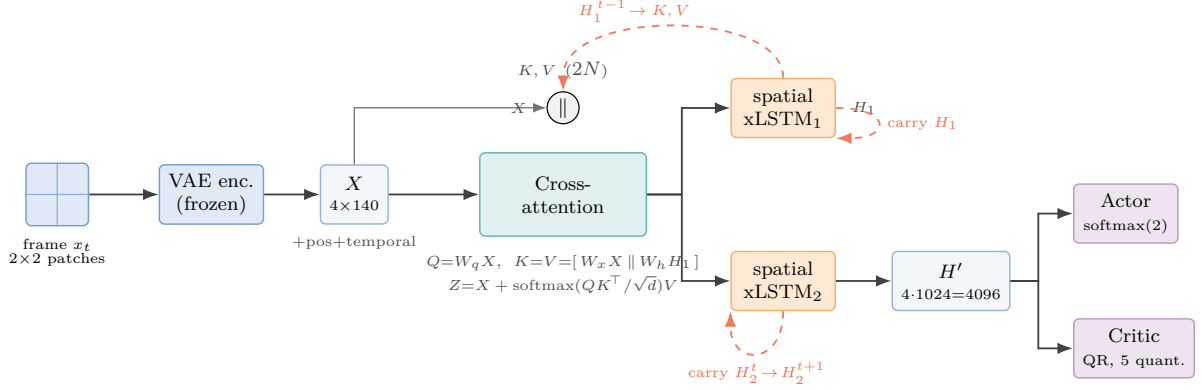
Multiplicative (paper-faithful) — behaviour (T=7, validity4)



**Figure A.6.** Multiplicative (paper-faithful reference) — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.

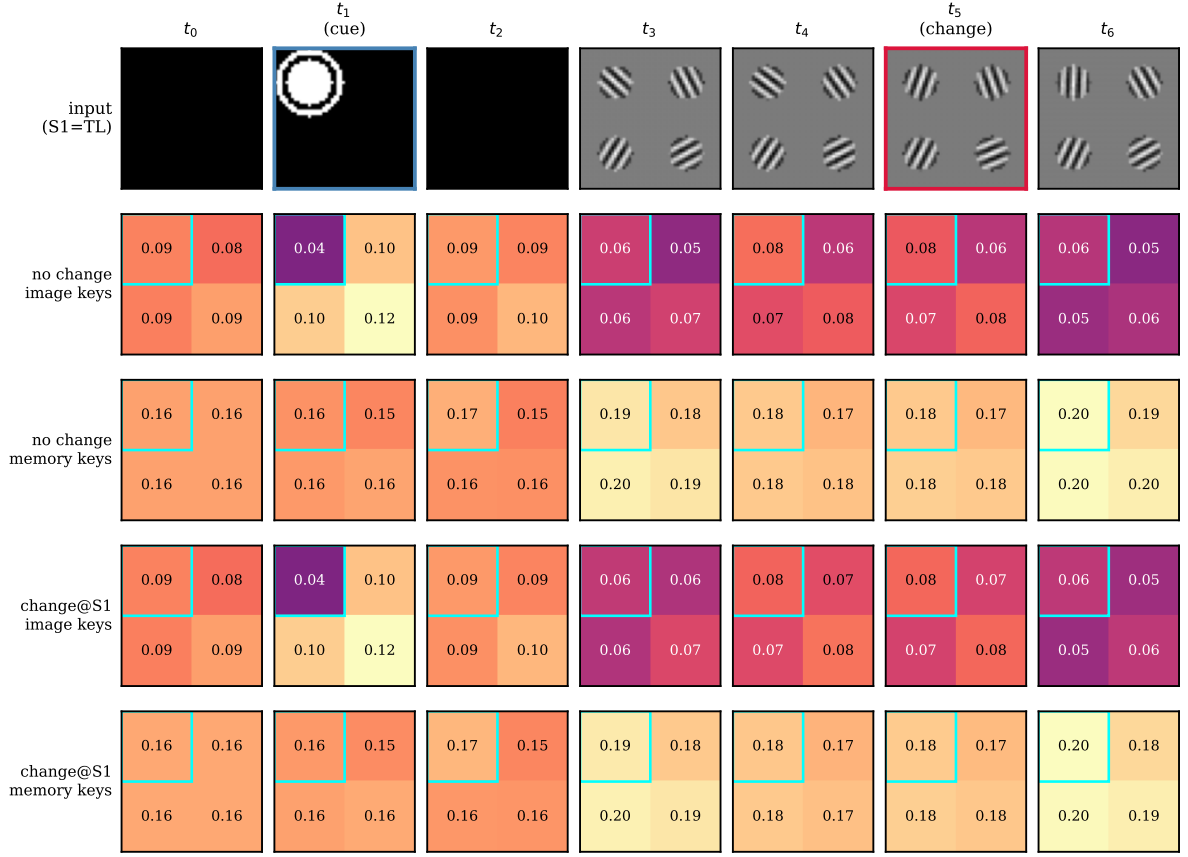
### A.3 Cross-attention (image + memory keys)

Works (mean cued-hit 0.70). The image queries keys/values drawn from both the image and the memory  $H_1$ . Its attention to the cued *image* patch sits *below* uniform ( $\sim 0.06$ – $0.09$ ) while the memory keys carry the bulk of the weight — the model routes the cued location through memory rather than orienting image attention to it.

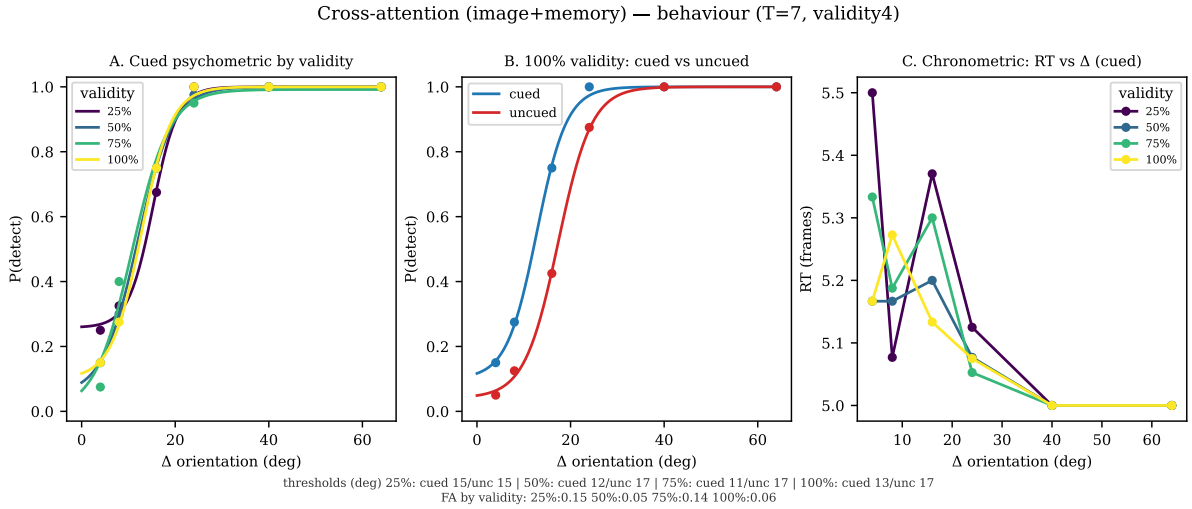


**Figure A.7. Cross-attention (image + memory keys)** — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.

Cross-attention (image+memory) — attention maps, 100% cue at  $S_1$  (cyan = cued/changed patch)



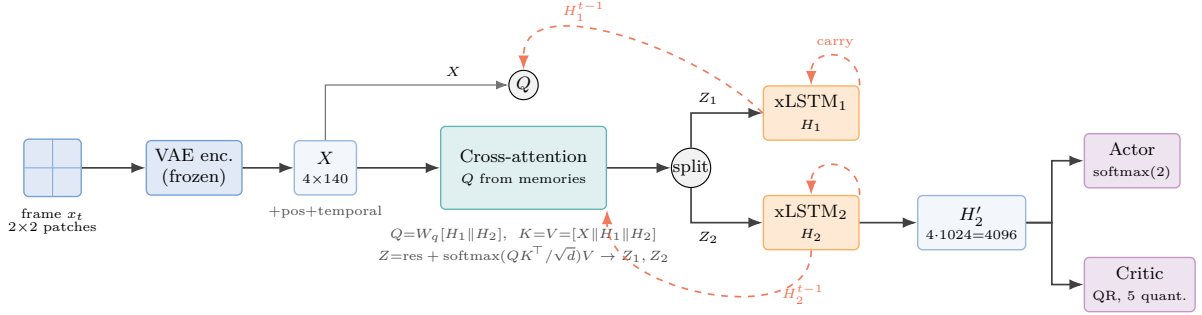
**Figure A.8.** Cross-attention (image + memory keys) — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).



**Figure A.9.** Cross-attention (image + memory keys) — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.

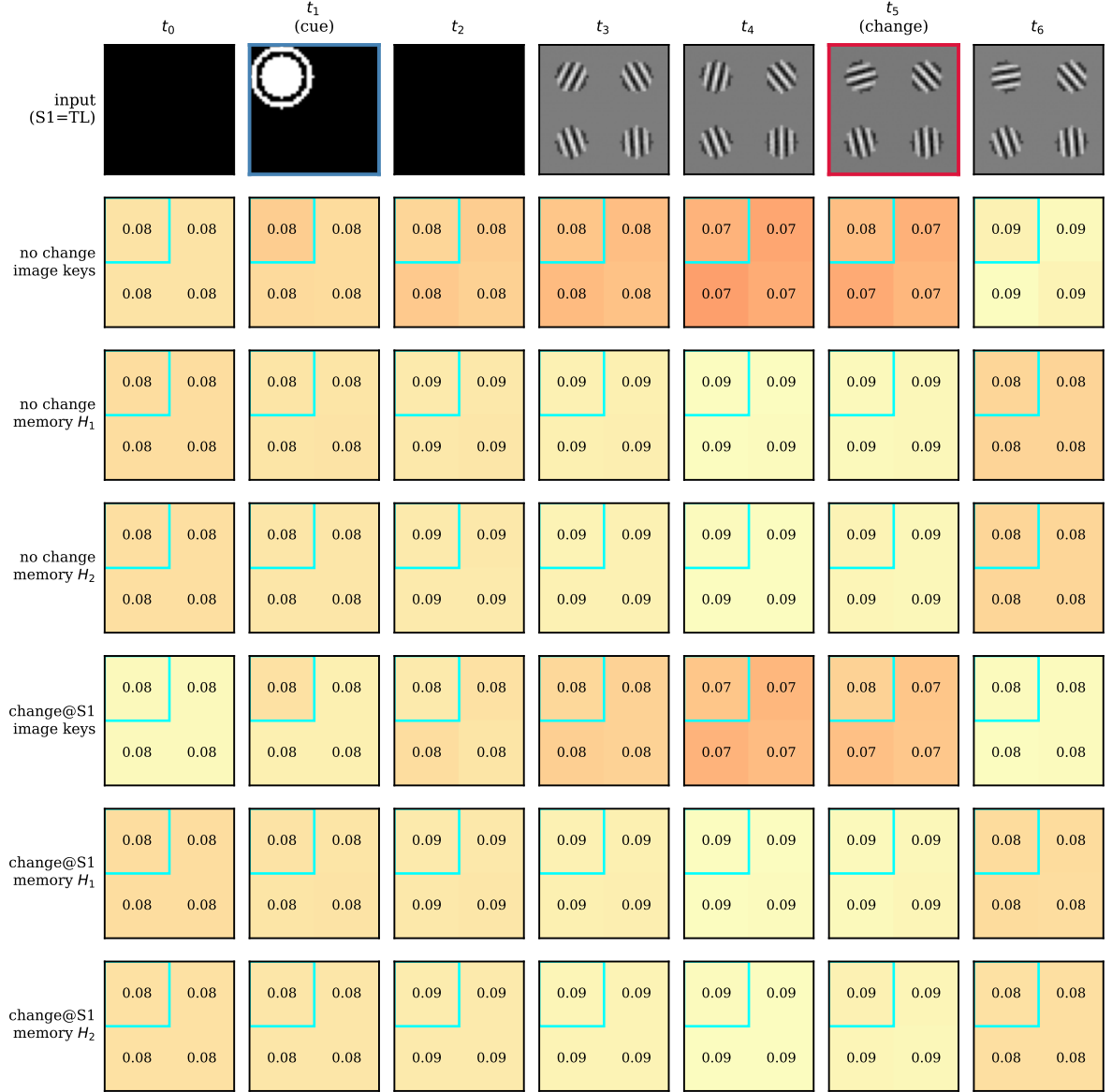
## A.4 Dual-memory (the memories query)

Works (mean cued-hit 0.64). Both memories form the queries; keys/values come from the image and both memories. The attention spreads near-uniformly across all twelve keys, and the image keys at the cued patch again sit below uniform.

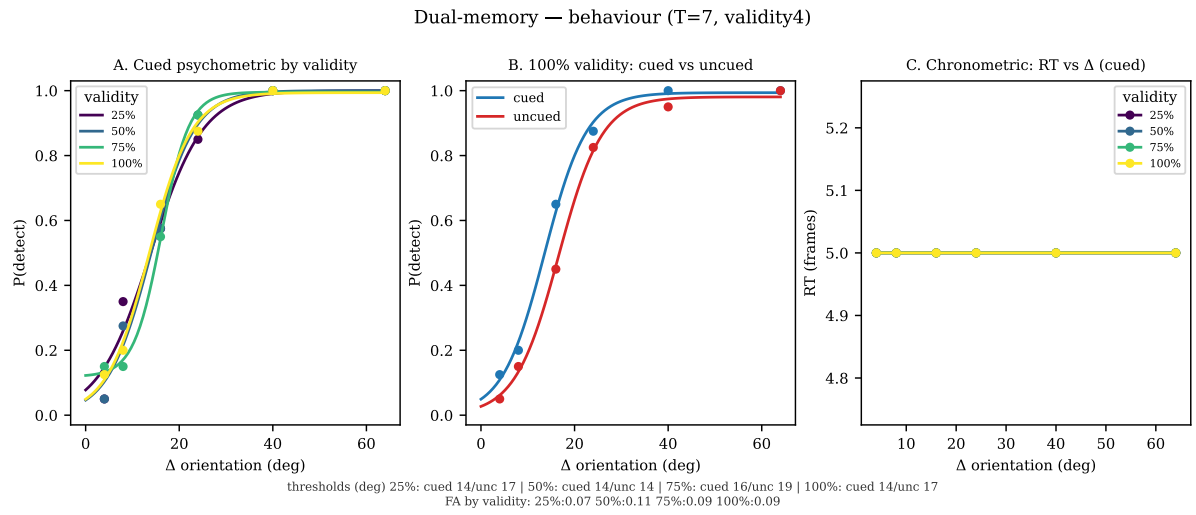


**Figure A.10.** Dual-memory (the memories query) — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.

Dual-memory — attention maps, 100% cue at  $S_1$  (cyan = cued/changed patch)



**Figure A.11. Dual-memory (the memories query)** — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).

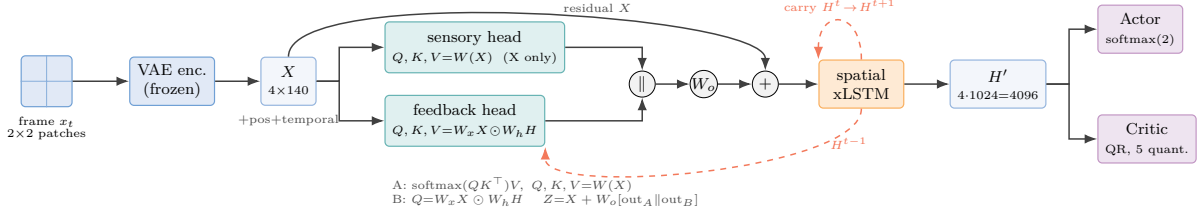


**Figure A.12. Dual-memory (the memories query) — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.**



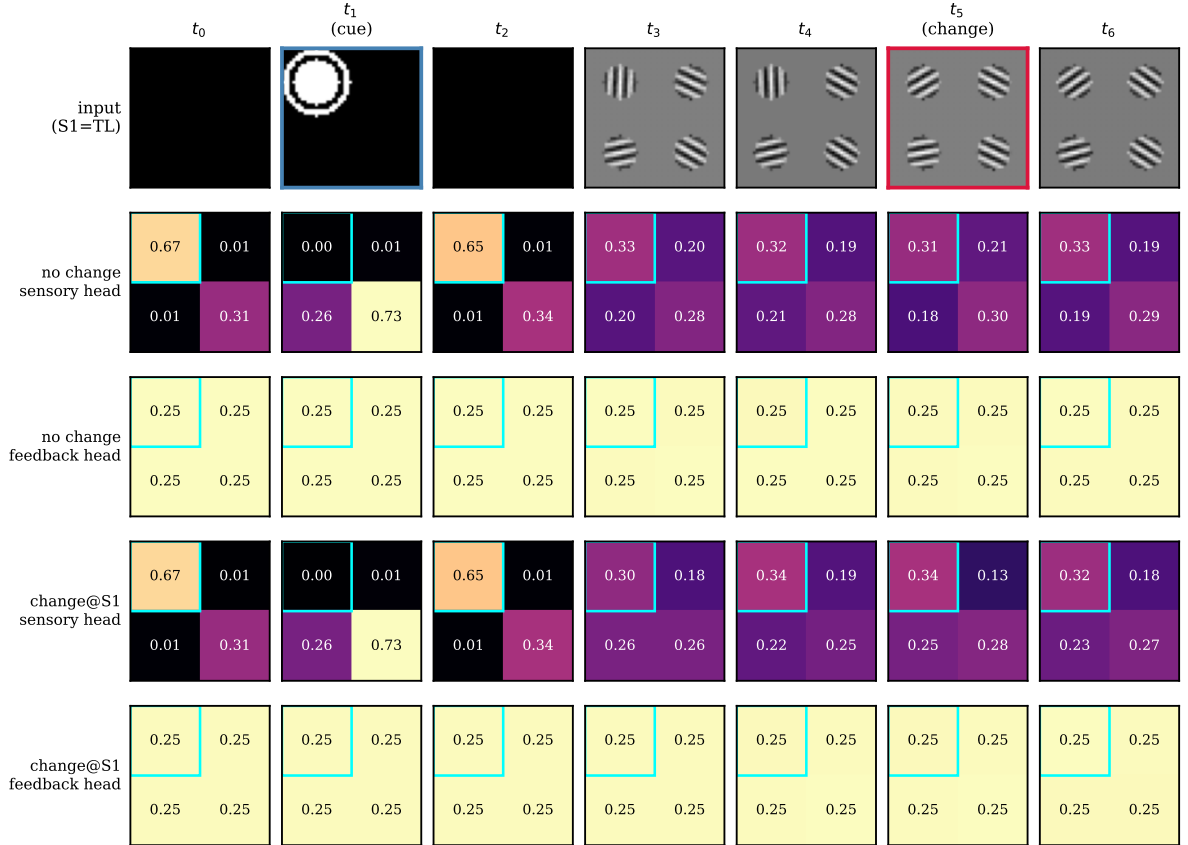
## A.5 Dual-head (sensory + feedback)

Works (mean cued-hit 0.63). The memory-free sensory head carries the family’s only *positive* cue-orienting bias (+0.04 above uniform) — real but far too weak to feature as an attention-allocation result.



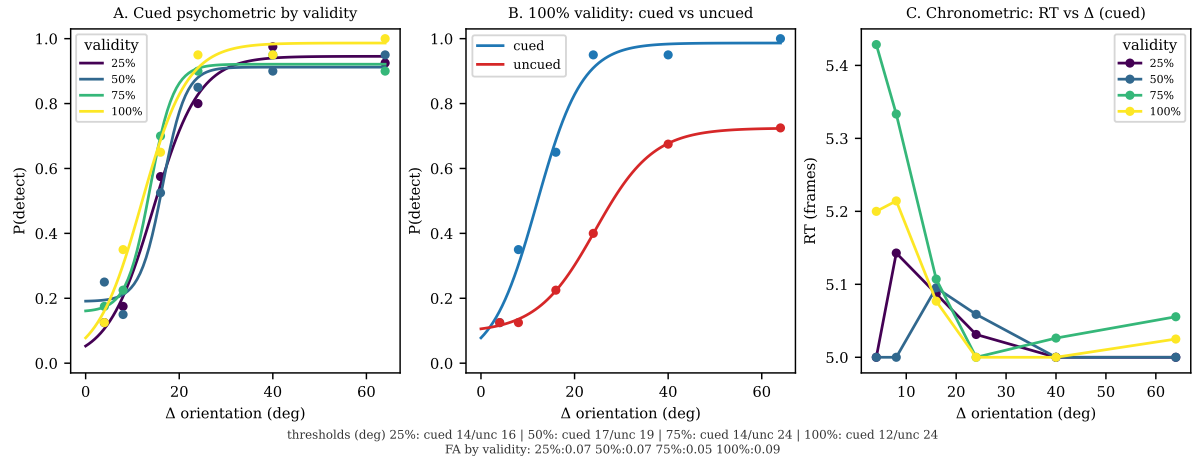
**Figure A.13.** Dual-head (sensory + feedback) — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.

Dual-head (sensory+feedback) — attention maps, 100% cue at S1 (cyan = cued/changed patch)



**Figure A.14.** Dual-head (sensory + feedback) — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).

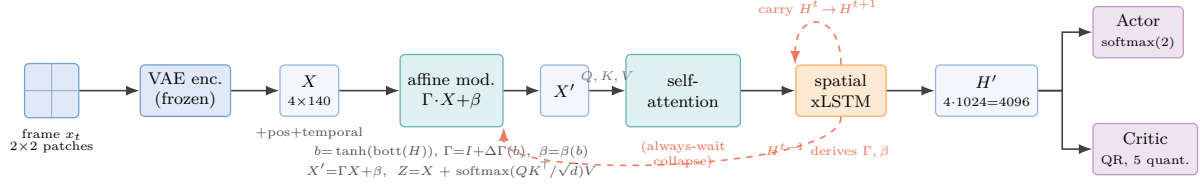
Dual-head (sensory+feedback) — behaviour (T=7, validity4)



**Figure A.15.** Dual-head (sensory + feedback) — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.

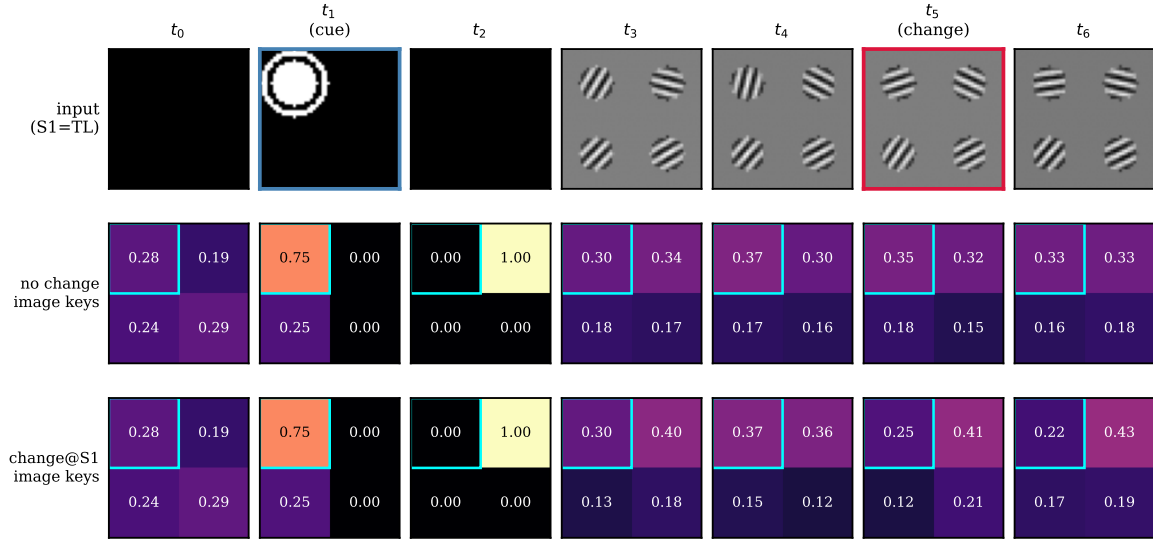
## A.6 Affine (matrix scale + shift) modulation — failed

**Failed to learn the task:** an always-wait collapse (never presses;  $P(\text{press}) \leq 0.004$  even on a clear 60° change). The architecture is shown for completeness; the behaviour panel is flat at zero by construction.

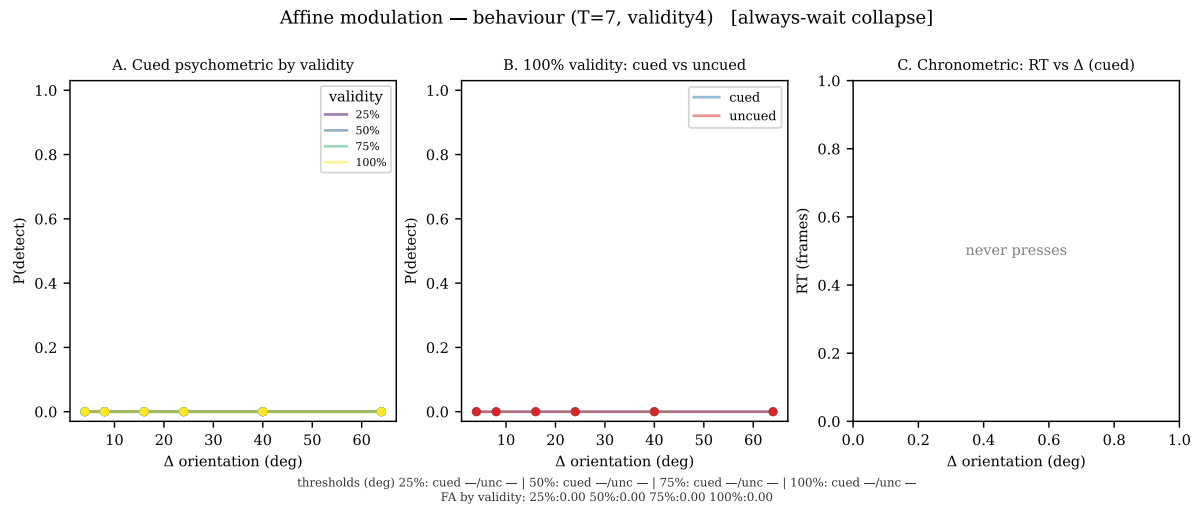


**Figure A.16.** Affine (matrix scale + shift) modulation — failed — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.

Affine modulation — attention maps, 100% cue at  $S_1$  (cyan = cued/changed patch) [always-wait collapse]



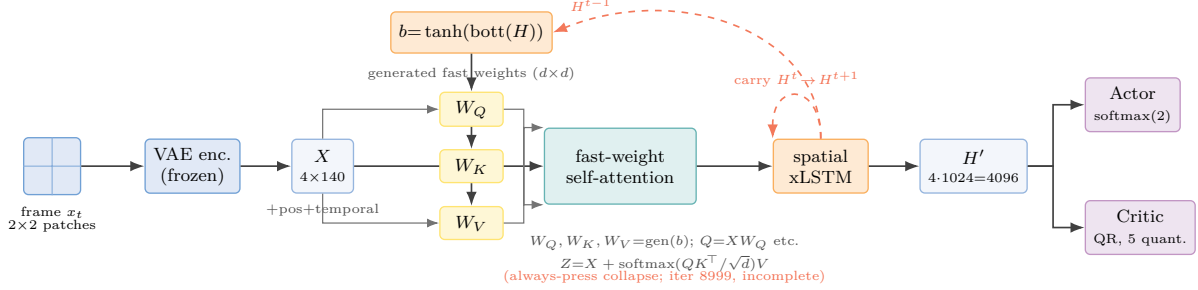
**Figure A.17.** Affine (matrix scale + shift) modulation — failed — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).



**Figure A.18. Affine (matrix scale + shift) modulation — failed — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.**

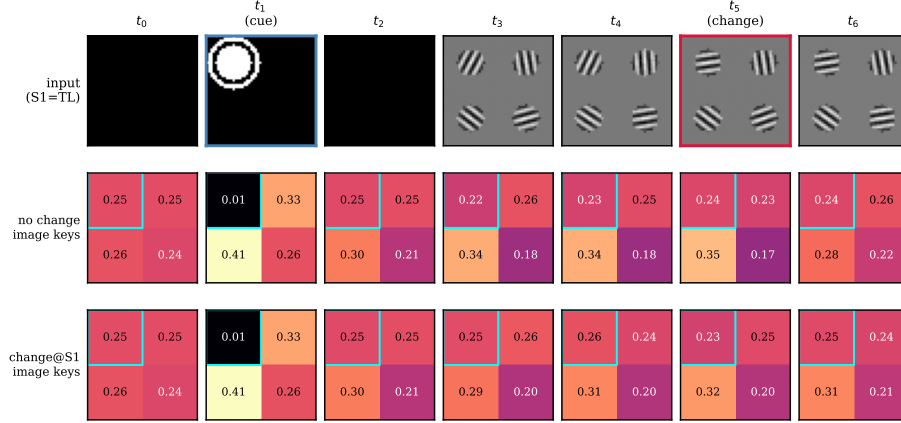
## A.7 Fast-weight (hypernetwork) attention — degenerate

Memory generates the  $Q/K/V$  projection weights. This checkpoint is in the *other* 0.5 collapse — always-press (hit 1.0, false-alarm 1.0) — and its training is incomplete (iter 8,999 of 18,750). Shown for completeness.

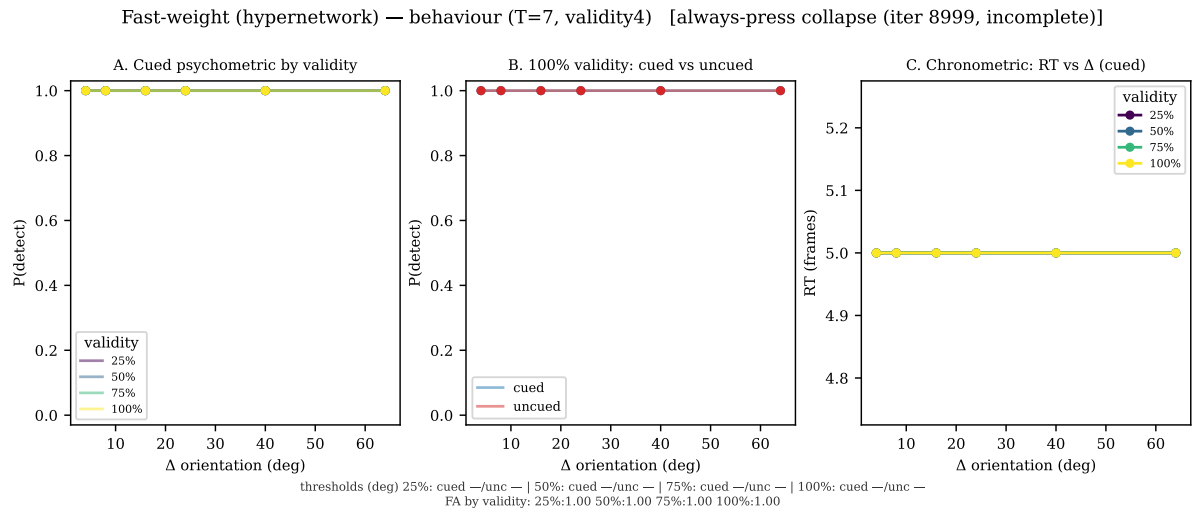


**Figure A.19. Fast-weight (hypernetwork) attention — degenerate — architecture.** Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.

Fast-weight (hypernetwork) — attention maps, 100% cue at  $S_1$  (cyan = cued/changed patch) [always-press collapse (iter 8999, incomplete)]



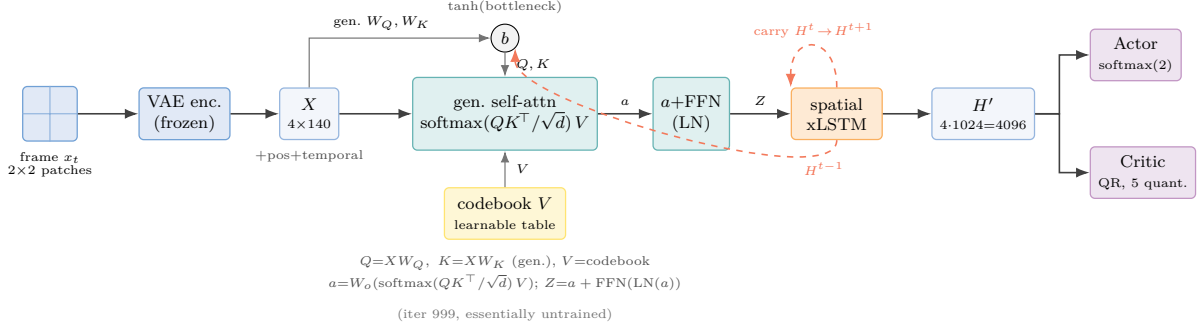
**Figure A.20. Fast-weight (hypernetwork) attention — degenerate — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).**



**Figure A.21. Fast-weight (hypernetwork) attention — degenerate — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.**

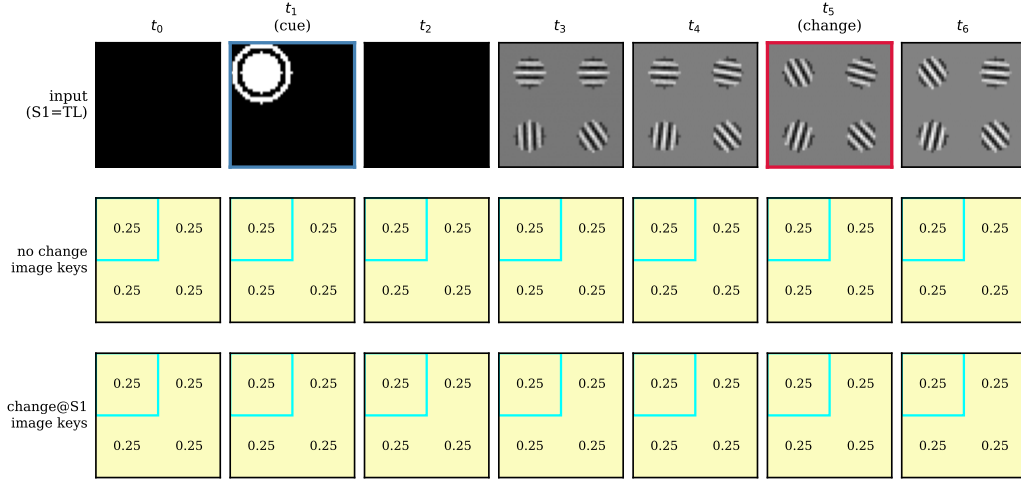
## A.8 Fast-weight + codebook — essentially untrained

Generated  $Q/K$  weights with a learnable codebook as the values and no image residual. This checkpoint is essentially untrained (iter 999) and sits in an always-wait collapse; included only so the catalogue is complete.

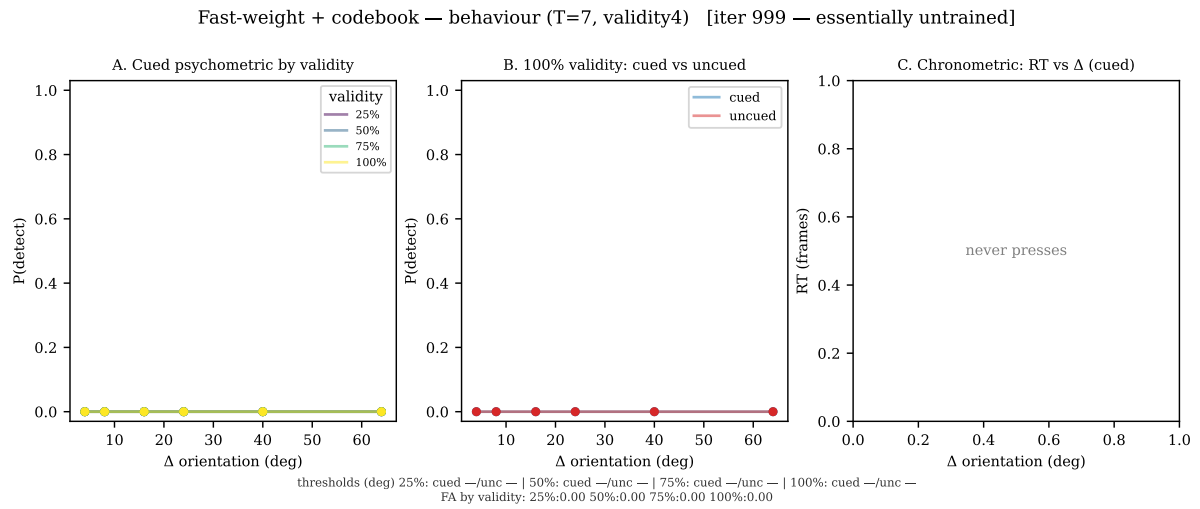


**Figure A.22.** Fast-weight + codebook — essentially untrained — architecture. Solid arrows: forward dataflow; dashed red: recurrent feedback from the previous frame.

Fast-weight + codebook — attention maps, 100% cue at  $S_1$  (cyan = cued/changed patch) [iter 999 — essentially untrained]



**Figure A.23.** Fast-weight + codebook — essentially untrained — attention maps at every frame, 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 150 trials).



**Figure A.24.** Fast-weight + codebook — essentially untrained — psychometric (A, B) and chronometric (C) functions on the seven-step validity4 task.



## Appendix B

# Per-model atlas: the successful $T = 29$ long-task models

This appendix gives the same per-model treatment as Appendix A, for the six models that trained successfully on the *long* ( $T = 29$ , random-onset) change-detection task: **v5**, **v5\_part2**, **v8**, **v11**, **v11\_part2**, and **v11\_part5**. These predate the paper-faithful build and live in separate codebases; they were each driven through one uniform analysis (the shared `_behav_utils` harness, or `distractor_dd` for **v11\_part5**) so the figures are directly comparable. Unlike the four-patch paper network, these tokenize the  $50 \times 50$  image into a  $10 \times 10$  grid (100 tokens), so the attention maps are  $10 \times 10$  and far richer.

**Two architectural families.** The *cross-attention* models (**v5**, **v5\_part2**, **v8**) place the image patches *and* the memories in the key set, so the attention has separate *image-key* and *memory-key* ( $H_1, H_2$ ) maps — both are shown. The *dual-stream* models (**v11**, **v11\_part2**, **v11\_part5**) use the image as the *query* and attend over memory only, so their maps are per-stream (salience  $\rightarrow H_1/H_2$ , top-down  $\rightarrow H_2$ ): they show *where in memory* the image query looks. **v11\_part5** was trained on a *cued-side* task with distractors; for comparability it is evaluated here on the standard cued task with distractors switched off.

**The headline, before the details.** On the long task these models *do* orient attention to the cue: the salience / image-key attention is biased toward the cued quadrant *before* the change (visible in panel D of each behaviour figure), and the reaction time is strongly graded with change magnitude (panel C,  $\sim 6 \rightarrow 1$  frames) rather than floored. This is the regime contrast anticipated by the short-task study (open question O5): the long horizon both permits and rewards pre-orienting, where the seven-step task did not.

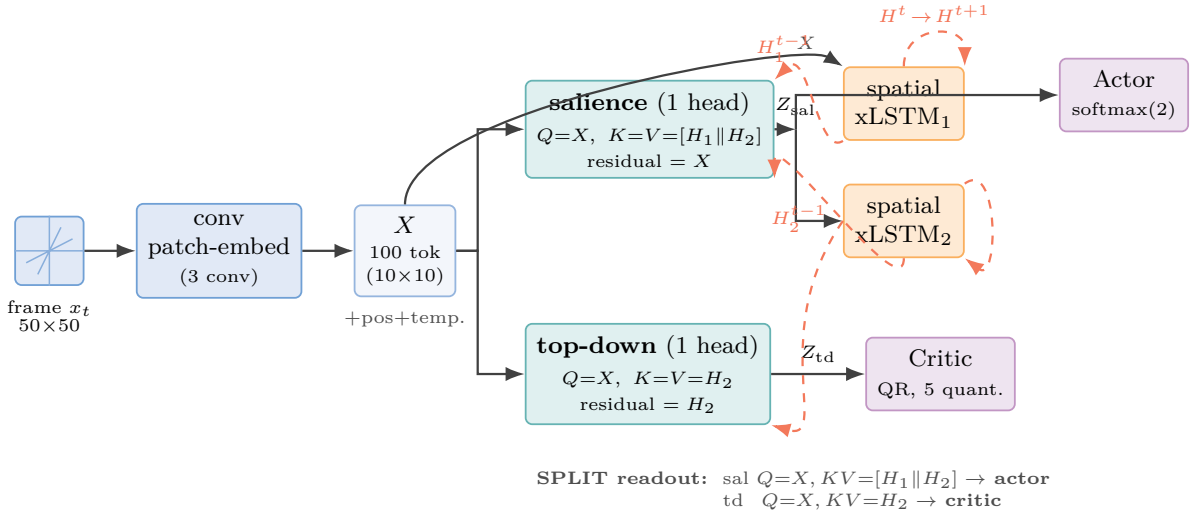
**Table B.1.** The six successful  $T = 29$  models. Mean cued-hit and false-alarm are averaged over the validity $\times$ magnitude psychometric grid (so the absolute hit is below the per-condition peak, which approaches 1.0 at large  $\Delta$ ). Snapshot iteration as analysed.

| Model            | Family                             | snapshot iter | mean cued-hit | mean FA |
|------------------|------------------------------------|---------------|---------------|---------|
| <b>v5</b>        | cross-attention (memory-as-tokens) | 1999          | 0.750         | 0.038   |
| <b>v5_part2</b>  | cross-attention over memory        | 4999          | 0.714         | 0.044   |
| <b>v8</b>        | cross-attention, $H_1$ residual    | 4999          | 0.721         | 0.019   |
| <b>v11</b>       | parallel dual-stream               | 8199          | 0.671         | 0.059   |
| <b>v11_part2</b> | split-readout cross-talk           | 10399         | 0.683         | 0.025   |
| <b>v11_part5</b> | split-readout (distractor task)    | 11799         | 0.748         | 0.022   |

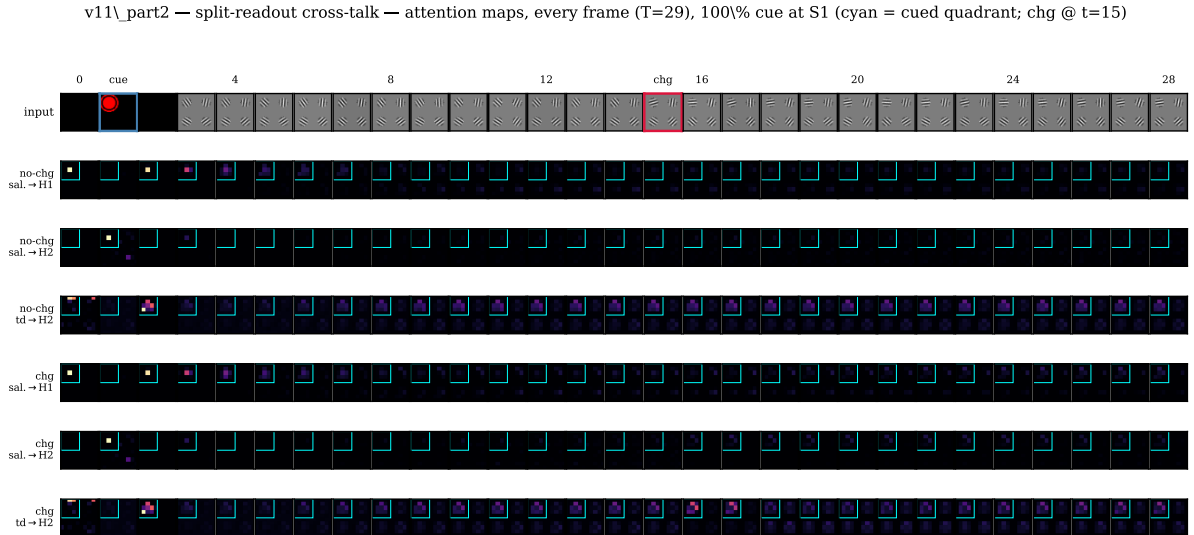


## B.1 v11\_part2 — split-readout cross-talk

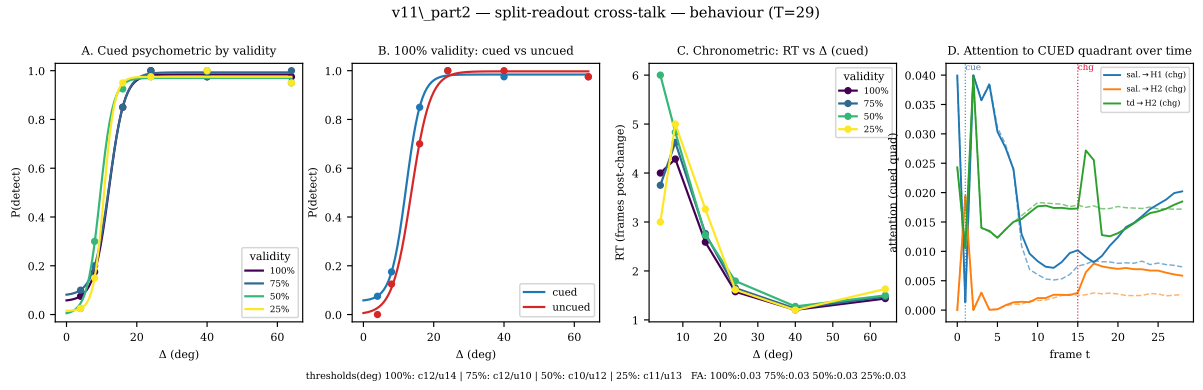
The flagship long-task model and the closest to the paper’s attention story. The salience stream ( $Q=X$ ,  $K=V=[H_1\|H_2]$ ) feeds the actor; the top-down stream ( $Q=X$ ,  $K=V=H_2$ ) feeds the critic. The cued-quadrant timecourse (panel D) shows the salience attention elevated at the cue and the top-down attention rising at the change; the reaction time is strongly graded. The cued psychometric sits left of the uncued (a genuine cueing benefit).



**Figure B.1.** v11\_part2 — split-readout cross-talk — architecture. Conv patch-embed  $\rightarrow 10 \times 10$  tokens; solid arrows forward, dashed red recurrent feedback.



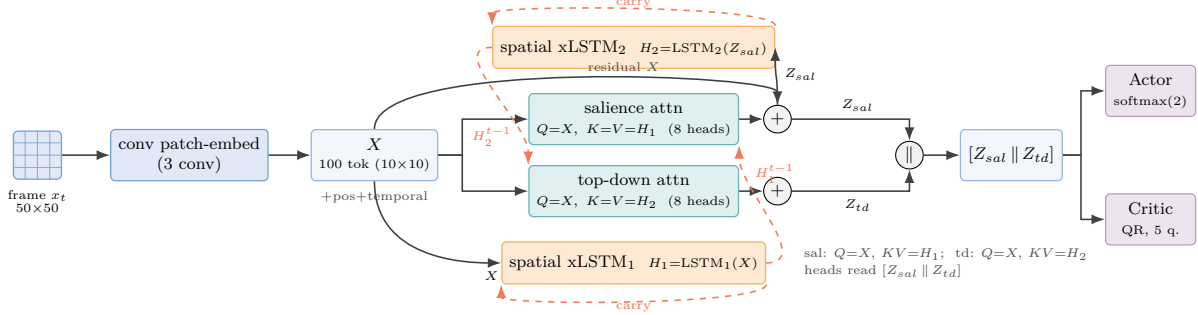
**Figure B.2.** v11\_part2 — split-readout cross-talk — attention maps at every frame ( $T = 29$ ), 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 80 trials; change at  $t = 15$ ).



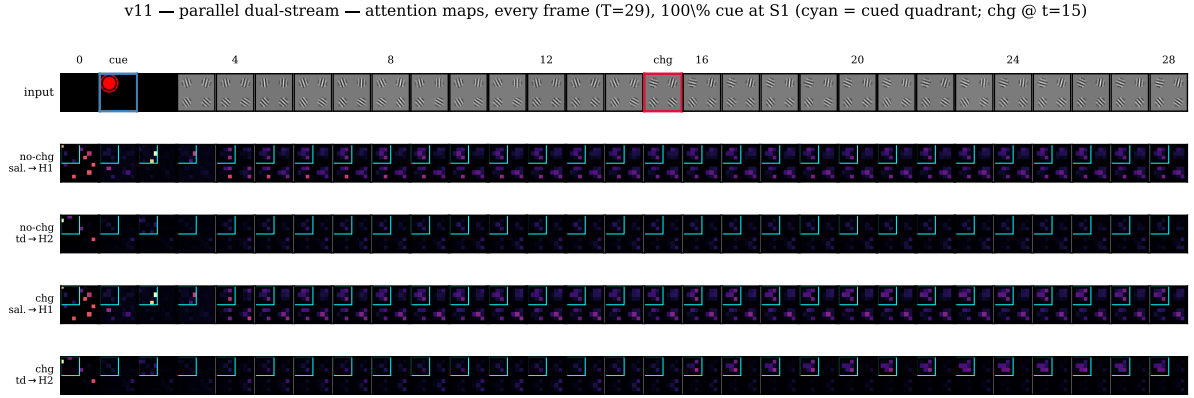
**Figure B.3.** v11\_part2 — split-readout cross-talk — psychometric (A,B), chronometric (C), and the cued-quadrant attention timecourse (D, the readable summary of the per-frame maps).

## B.2 v11 — parallel dual-stream

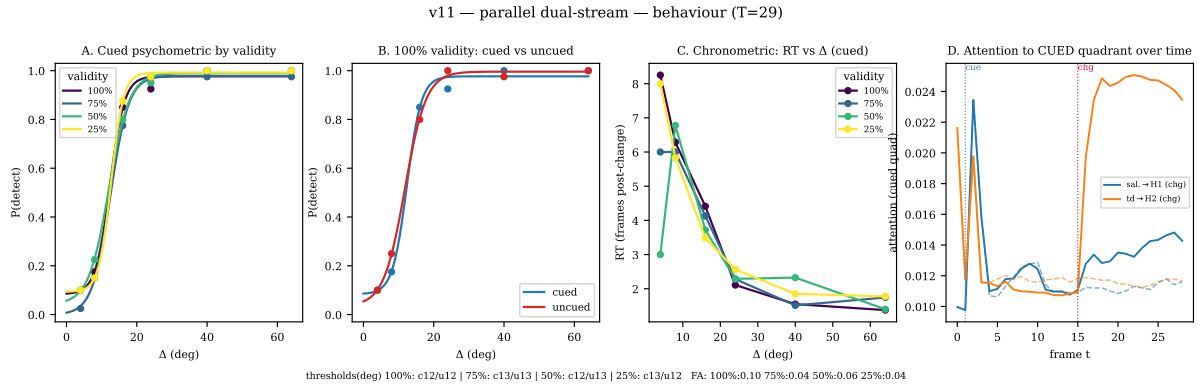
Two independent streams sharing only the image: salience ( $Q=X$ ,  $K=V=H_1$ , grounded by an  $X$  residual) and top-down ( $Q=X$ ,  $K=V=H_2$ ); the heads read both transformer outputs. The salience stream carries the cue prior, the top-down stream the change-lock.



**Figure B.4.** v11 — parallel dual-stream — architecture. Conv patch-embed  $\rightarrow 10 \times 10$  tokens; solid arrows forward, dashed red recurrent feedback.



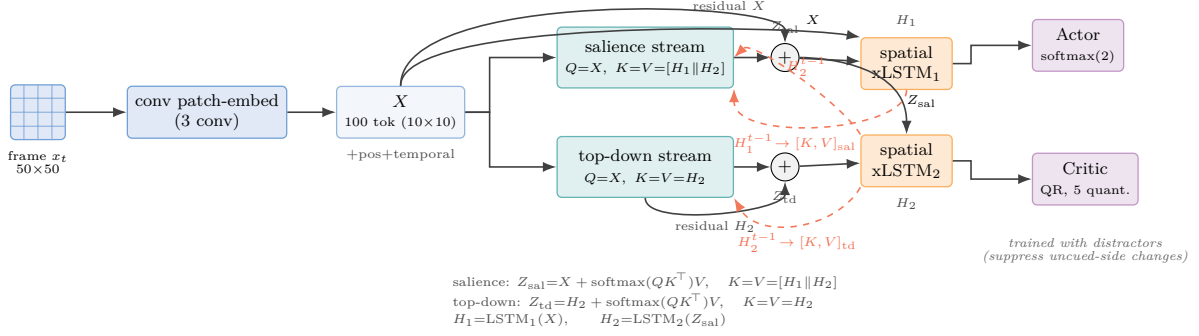
**Figure B.5.** v11 — parallel dual-stream — attention maps at every frame ( $T = 29$ ), 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 80 trials; change at  $t = 15$ ).



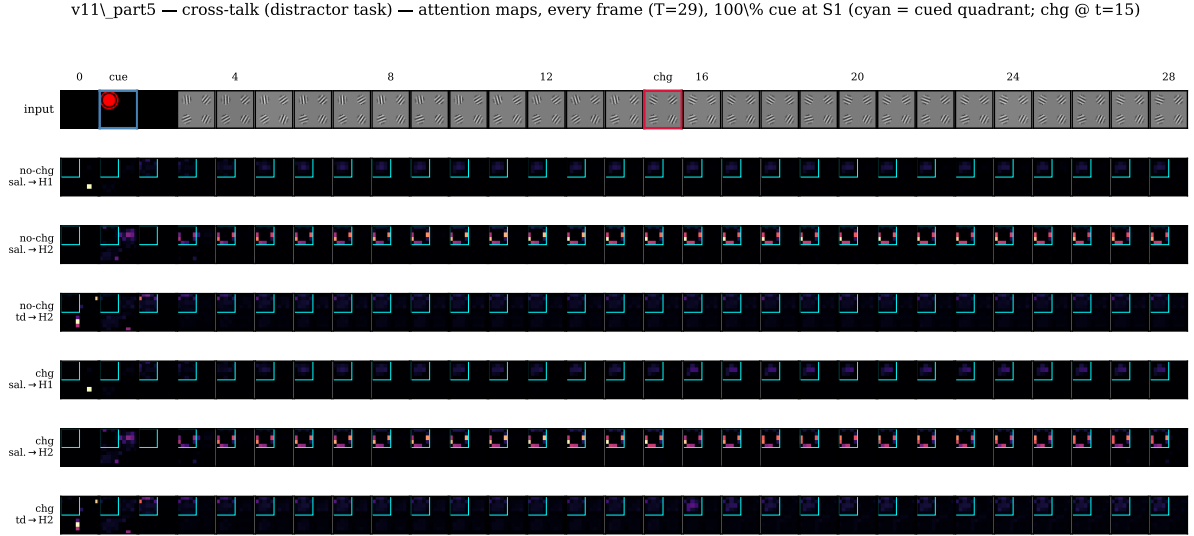
**Figure B.6.** v11 — parallel dual-stream — psychometric (A,B), chronometric (C), and the cued-quadrant attention timecourse (D, the readable summary of the per-frame maps).

### B.3 v11\_part5 — split-readout cross-talk (distractor-trained)

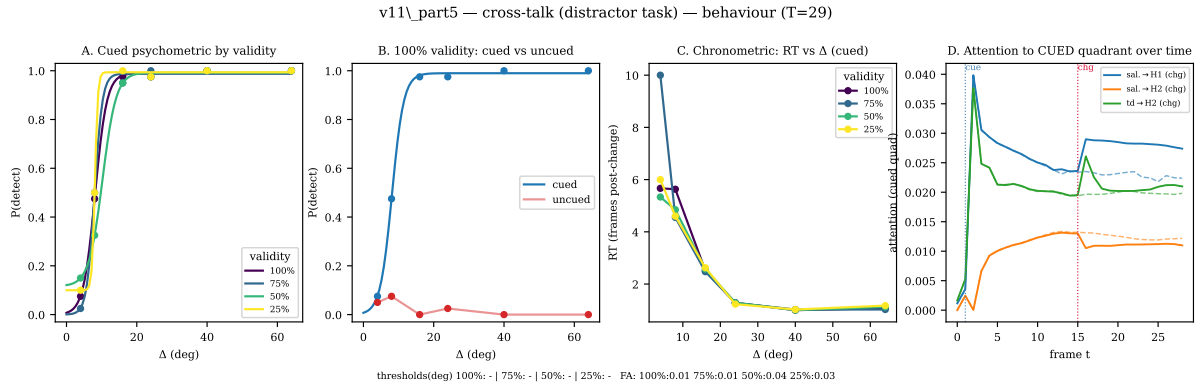
Architecturally identical to v11\_part2 but trained on the cued-*side* task with distractors (the uncued side changes on half the trials and must be suppressed). Evaluated here on the standard cued task with distractors off, it is among the strongest performers (mean cued-hit 0.75).



**Figure B.7.** v11\_part5 — split-readout cross-talk (distractor-trained) — architecture. Conv patch-embed  $\rightarrow 10 \times 10$  tokens; solid arrows forward, dashed red recurrent feedback.



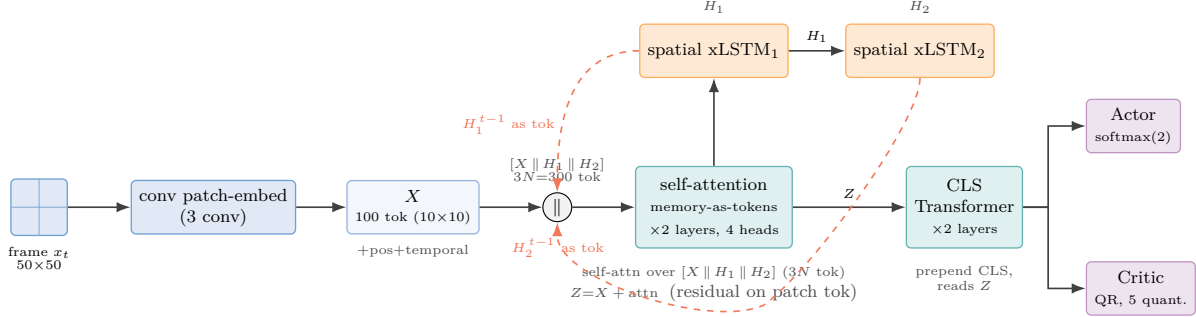
**Figure B.8.** v11\_part5 — split-readout cross-talk (distractor-trained) — attention maps at every frame ( $T = 29$ ), 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 80 trials; change at  $t = 15$ ).



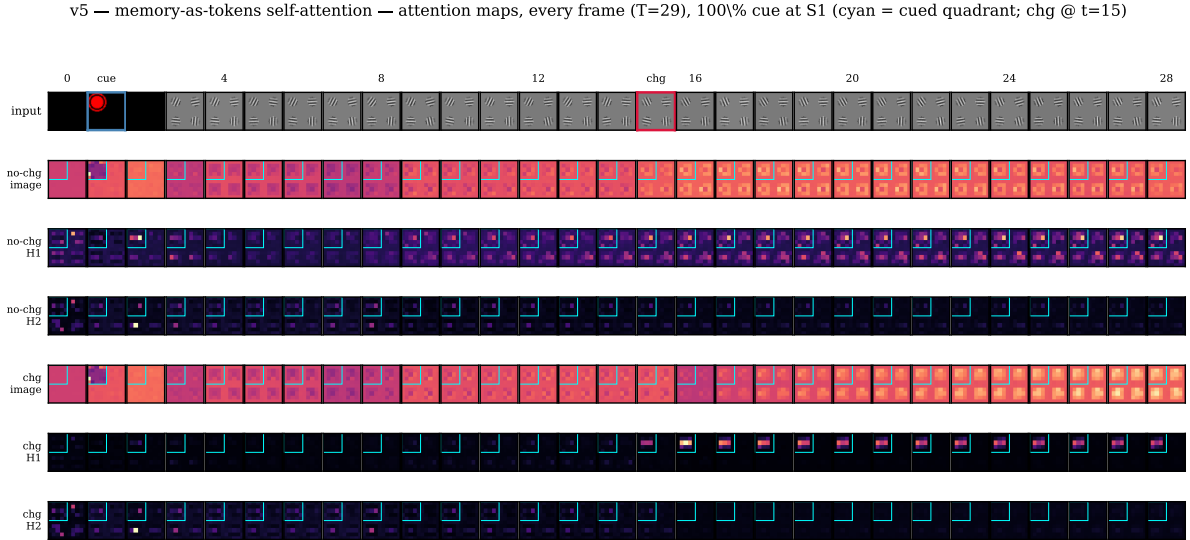
**Figure B.9.** v11\_part5 — split-readout cross-talk (distractor-trained) — psychometric (A,B), chronometric (C), and the cued-quadrant attention timecourse (D, the readable summary of the per-frame maps).

## B.4 v5 — memory-as-tokens self-attention

Each encoder layer self-attends over the concatenation  $[X \parallel H_1 \parallel H_2]$  (300 tokens). The image-key map (panel rows “image”) tiles the four Gabor quadrants as they appear, and a memory hotspot tracks the change; the first encoder layer acts as a cue-driven spatial orienter.

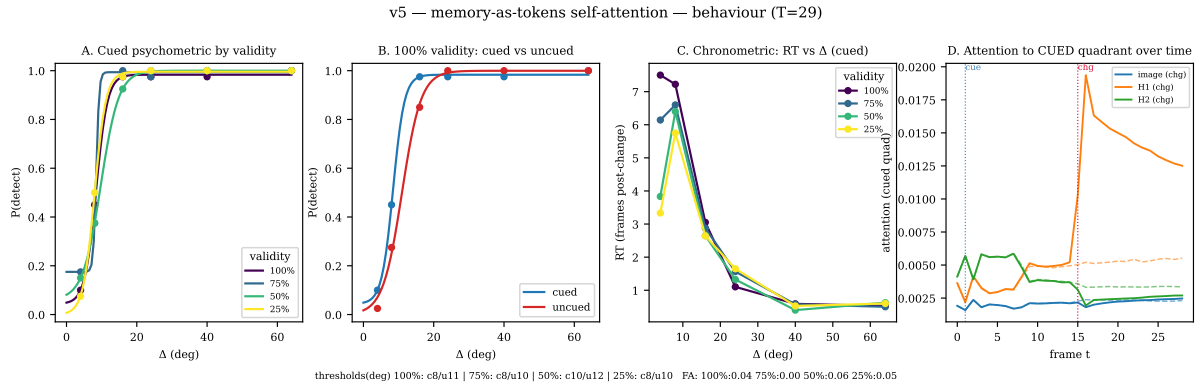


**Figure B.10.** v5 — memory-as-tokens self-attention — architecture. Conv patch-embed  $\rightarrow 10 \times 10$  tokens; solid arrows forward, dashed red recurrent feedback.



**Figure B.11.** v5 — memory-as-tokens self-attention — attention maps at every frame ( $T = 29$ ), 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 80 trials; change at  $t = 15$ ).

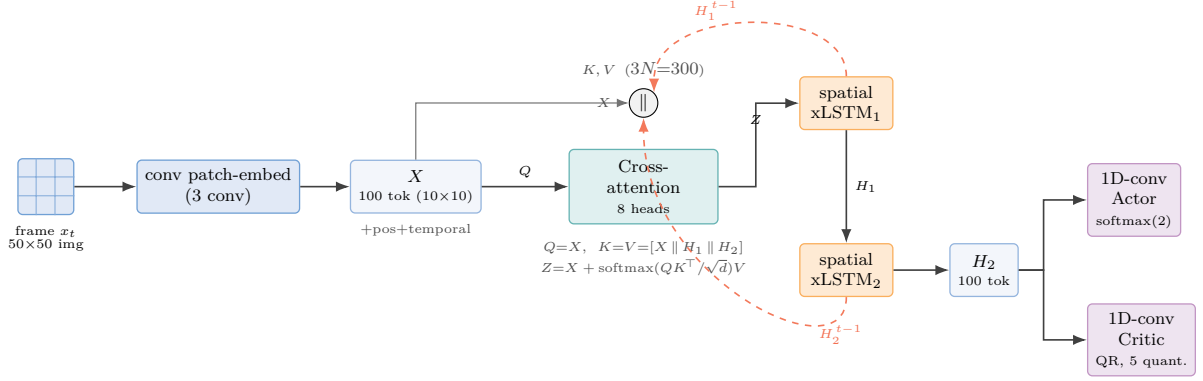




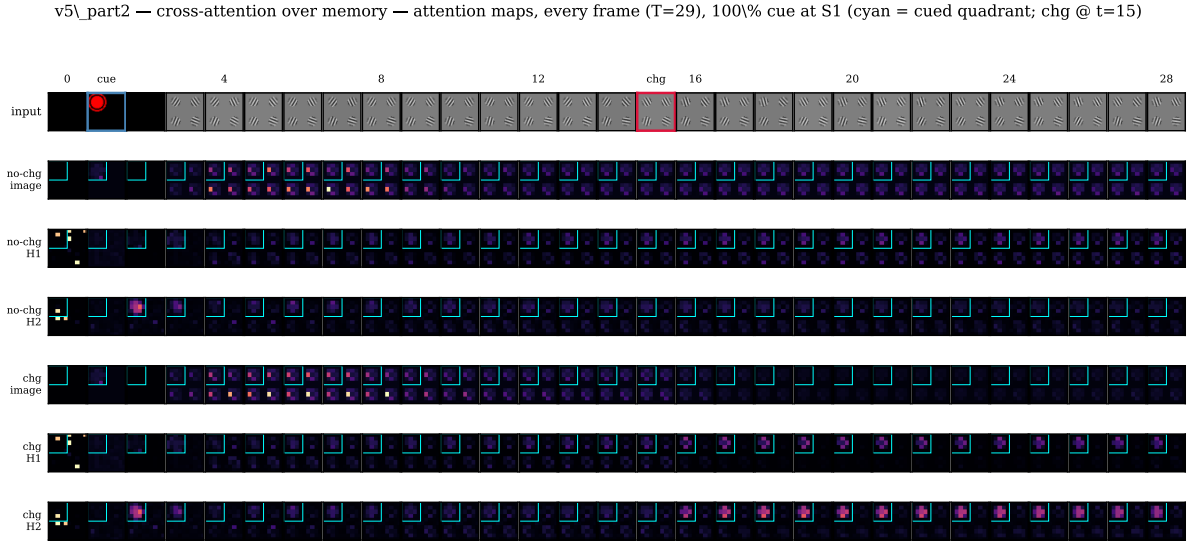
**Figure B.12. v5 — memory-as-tokens self-attention** — psychometric (A,B), chronometric (C), and the cued-quadrant attention timecourse (D, the readable summary of the per-frame maps).

## B.5 v5\_part2 — cross-attention over memory

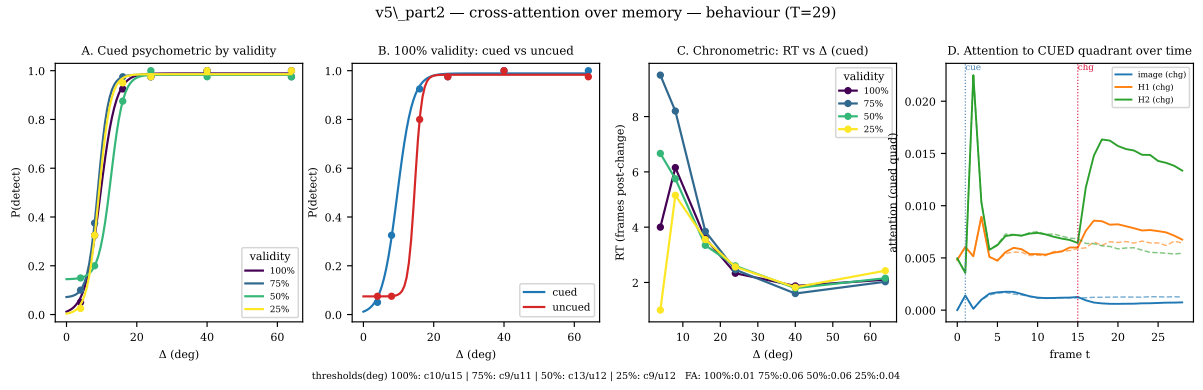
Cross-attention with  $Q=X$ ,  $K=V=[X \parallel H_1 \parallel H_2]$  and two stacked LSTMs read by 1D-conv heads. The load-bearing computation is in the memory cross-attention: the memory-key maps carry the decision-relevant structure while the image keys provide the spatial substrate.



**Figure B.13.** v5\_part2 — cross-attention over memory — architecture. Conv patch-embed  $\rightarrow 10 \times 10$  tokens; solid arrows forward, dashed red recurrent feedback.



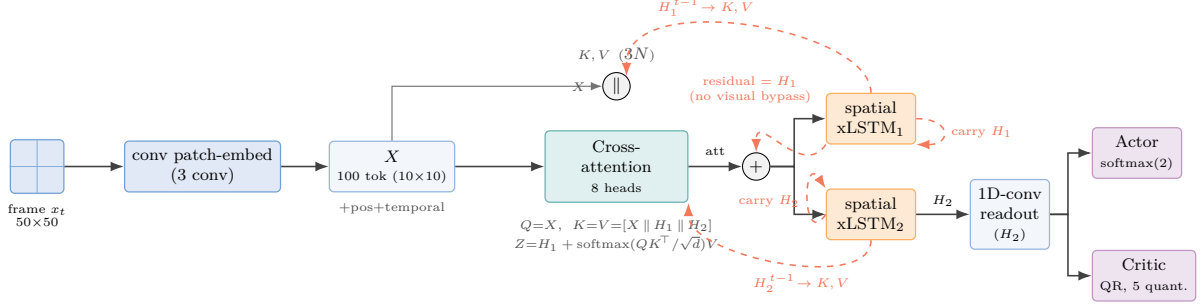
**Figure B.14.** v5\_part2 — cross-attention over memory — attention maps at every frame ( $T = 29$ ), 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 80 trials; change at  $t = 15$ ).



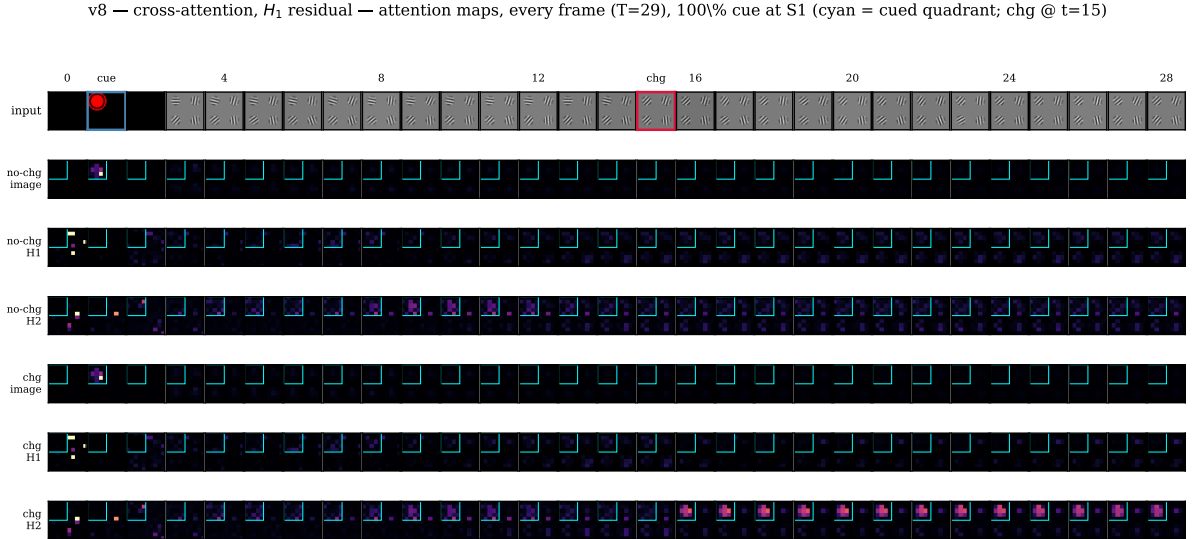
**Figure B.15.** v5\_part2 — cross-attention over memory — psychometric (A,B), chronometric (C), and the cued-quadrant attention timecourse (D, the readable summary of the per-frame maps).

## B.6 v8 — cross-attention with an $H_1$ residual

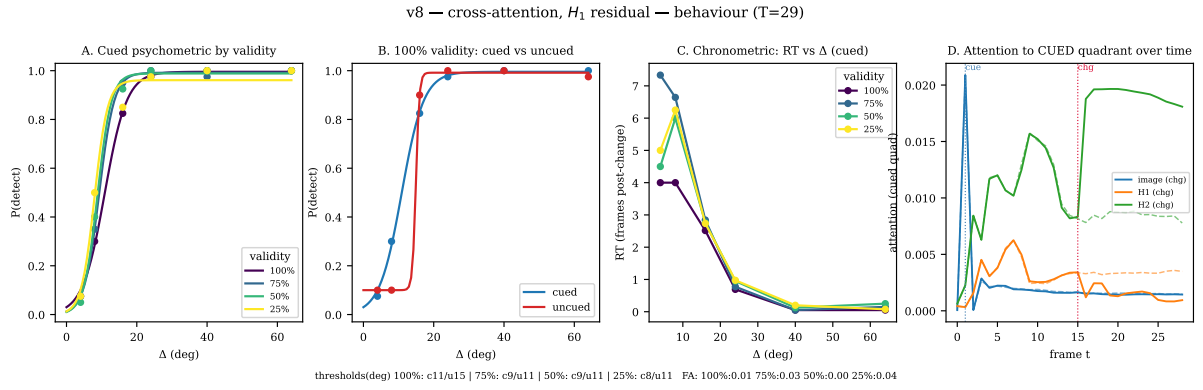
v5\_part2 with the cross-attention residual set to the previous-frame memory  $H_1$  instead of the image, so visual information can reach the decision *only* through the attended values. A dedicated patch-reading head emerges and reaction times are fast; the image pathway becomes co-dominant with memory.



**Figure B.16.** v8 — cross-attention with an  $H_1$  residual — architecture. Conv patch-embed  $\rightarrow 10 \times 10$  tokens; solid arrows forward, dashed red recurrent feedback.



**Figure B.17.** v8 — cross-attention with an  $H_1$  residual — attention maps at every frame ( $T = 29$ ), 100% cue at  $S_1$ , for the no-change and change-at- $S_1$  conditions (mean of 80 trials; change at  $t = 15$ ).



**Figure B.18.** v8 — cross-attention with an  $H_1$  residual — psychometric (A,B), chronometric (C), and the cued-quadrant attention timecourse (D, the readable summary of the per-frame maps).